

INDUSTRY'S NEXT LEAP: STRUCTURAL TRANSFORMATION AND GLOBAL INTEGRATION



The industrial sector in FY 2025-26 continued to display resilience and further regained momentum despite persistent global headwinds. This momentum is underpinned partly by the shift towards high-technology manufacturing. This transition reflects global trends wherein competitiveness hinges less on cost arbitrage and more on strategic indispensability within Global Value Chains (GVCs). India's long-term success will depend on its ability to integrate into GVCs as a high-tech, high-productivity manufacturing hub supported by stable and innovation-friendly policies.

Flagship policy initiatives of the Government are anchoring this transformation. The Production Linked Incentive (PLI) scheme continues to attract substantial investments and drive domestic value addition, particularly in electronics and emerging technology segments. Complementing this, the National Manufacturing Mission sets a long-term blueprint to boost the manufacturing sector substantially. A critical focus is on transitioning India from a technology adopter to a global innovator and scaling up the MSMEs for global competitiveness. The innovation ecosystem is being strengthened through the establishment of the Anusandhan National Research Foundation (ANRF) and the Research, Development & Innovation (RDI) Fund, which aim to crowd-in private R&D and accelerate leadership in strategic sectors such as semiconductors. Parallely, a strong thrust on infrastructure and logistics efficiency—through initiatives like PM GatiShakti and National Industrial Corridor Development Programme—is contributing towards enhancing cost competitiveness.

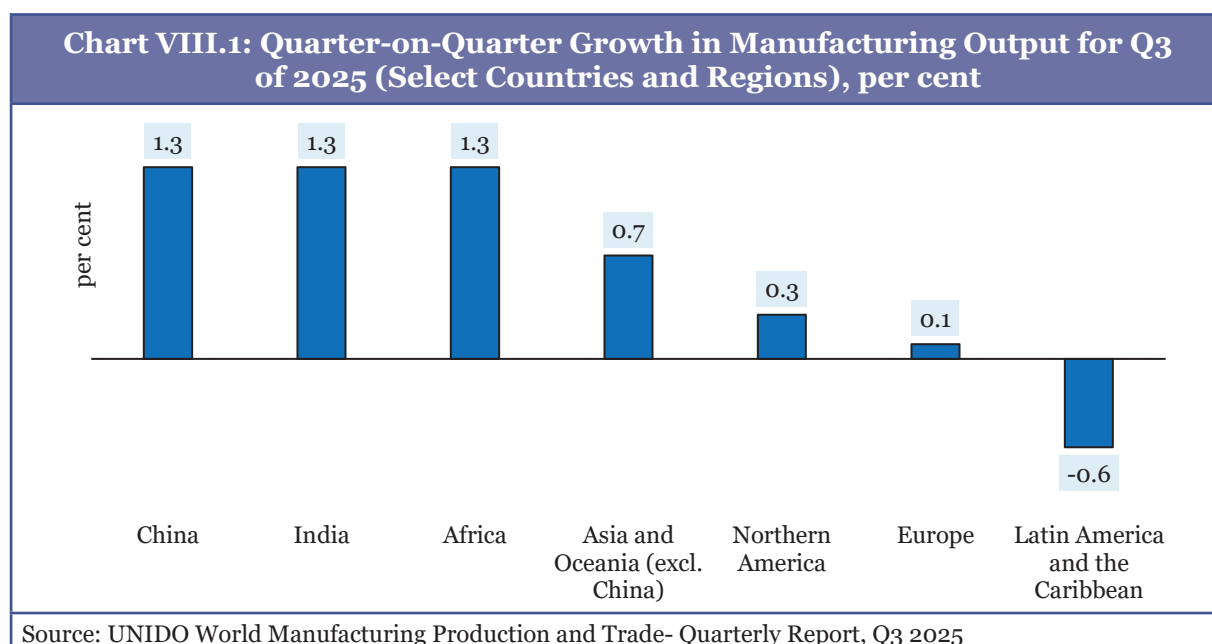
The path to Viksit Bharat @2047 requires strategic resilience and continuous ascent within the GVCs. Sustained reforms across five pillars- Ease of Doing Business, R&D and innovation, Skilling, Infrastructure & Logistics, and Scaling up of MSMEs- will remain critical in positioning industry as a key engine of future growth.

GLOBAL MANUFACTURING: UNEVEN RECOVERY AND THE SHIFT TOWARDS HIGH-TECH VALUE CHAINS

8.1. Globally, the industrial sector in 2025 continued to face macroeconomic challenges stemming from evolving geopolitics, inflationary pressures and supply

chain realignments among other factors. Despite a steady first half, the global outlook remains subdued, with risks tilted to the downside.¹

8.2. Though the world manufacturing output as a whole, expanded by 0.7 per cent in Q3 of 2025 (calendar year), the performance across regions remained uneven. Africa led manufacturing growth in Q3 2025, followed by Asia and Oceania. While Latin America and the Caribbean witnessed a decline, the Europe experienced stagnation.² India maintained strong industrial growth during the same period, placing it among the fastest growing industrial performers globally.



8.3. Global manufacturing continues to shift towards higher-value, innovation-intensive activities. Medium- and high-technology industries expanded by 1.7 per cent and 1.4 per cent in Q2 and Q3 of 2025 respectively, significantly outpacing stagnant low-tech segments.³ The UNCTAD World Investment Report 2025⁴ notes, '*investment in the digital economy and technology continues to act as a growth engine.*'

8.4. Alongside this technological shift, countries are also recalibrating their industrial strategies to reduce vulnerability to external disruption and enhance their leverage within global production networks by securing reliable positions in key areas of the global value chain.

¹ International Monetary Fund's World Economic Outlook for October 2025.

² UNIDO World Manufacturing Production and Trade: Q3 2025 Report available at <https://tinyurl.com/wde6xrsx>.

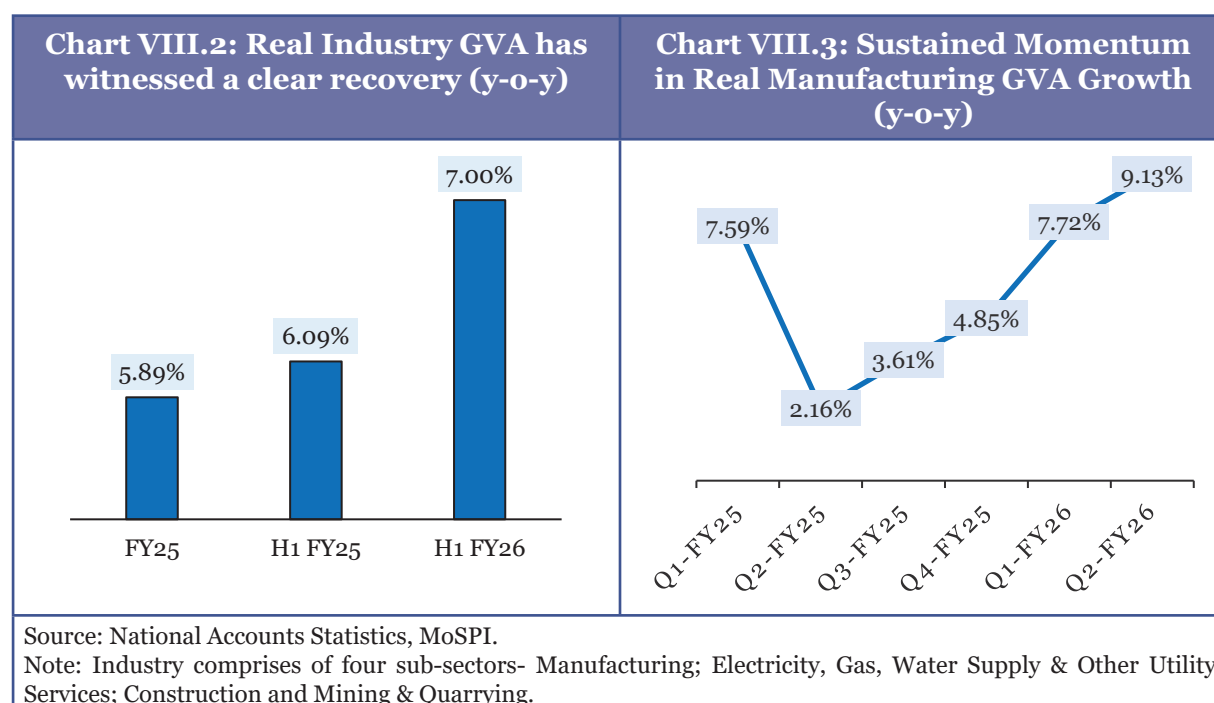
³ Ibid.

⁴ <https://tinyurl.com/5x52442e>.

8.5. Collectively, these trends suggest that the next phase of global manufacturing will be shaped less by simple cost arbitrage and more by strategic indispensability. Future success will depend on a country's preparedness and capacity to embed itself into GVCs as a high-tech, highly productive manufacturing hub supported by a stable policy environment.

DOMESTIC DEVELOPMENTS: RESILIENCE AND STRUCTURAL TRANSFORMATION

8.6. Against this backdrop, India's industrial performance remains robust. Industry Gross Value Added (GVA) grew by 7.0 per cent year-on-year, in real terms, in the first half of FY2025-26, marking a clear pickup after growth had eased to 5.9 per cent in the previous fiscal year (FY2024-25).



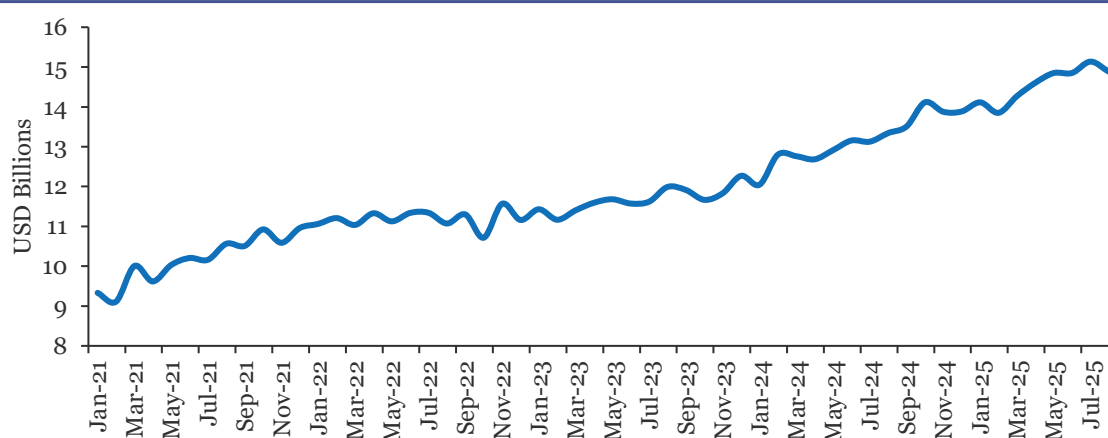
8.7. Worldwide, manufacturing activity softened across major economies through 2024 and early 2025, with some advanced economies even experiencing stagnation. This global deceleration dampened trade, investment, and order books everywhere, creating spillovers that inevitably impacted the Indian economy. Consequently, the easing in FY25 reflected a broader cooling of global demand rather than a weakening of India's underlying industrial capabilities.

8.8. This is further reinforced by the recovery witnessed in FY 2025-26, as Manufacturing GVA grew by 7.72 per cent and 9.13 per cent in Q1 and Q2, respectively, primarily driven by ongoing structural shifts within the sector. These include a gradual

move toward higher-value manufacturing segments, improved availability of industrial infrastructure through corridor-led development, and greater adoption of technology and formalisation across firms. This reflects India's improved industrial capability.

8.9. Medium and high-technology activities now account for 46.3 per cent of India's total manufacturing value added.⁵ This positions India among a relatively small group of middle-income industrialising economies that are steadily moving towards sophisticated production structures. This shift is due to various government initiatives such as the Production Linked Incentive (PLI) schemes and the India Semiconductor Mission, alongside the strengthening of domestic capacities in electronics, pharmaceuticals, chemicals and transportation sectors. Consequently, India's global standing has strengthened, as its ranking in terms of Competitive Industrial Performance (CIP) improved to 37th in 2023, up from 40th position in 2022.

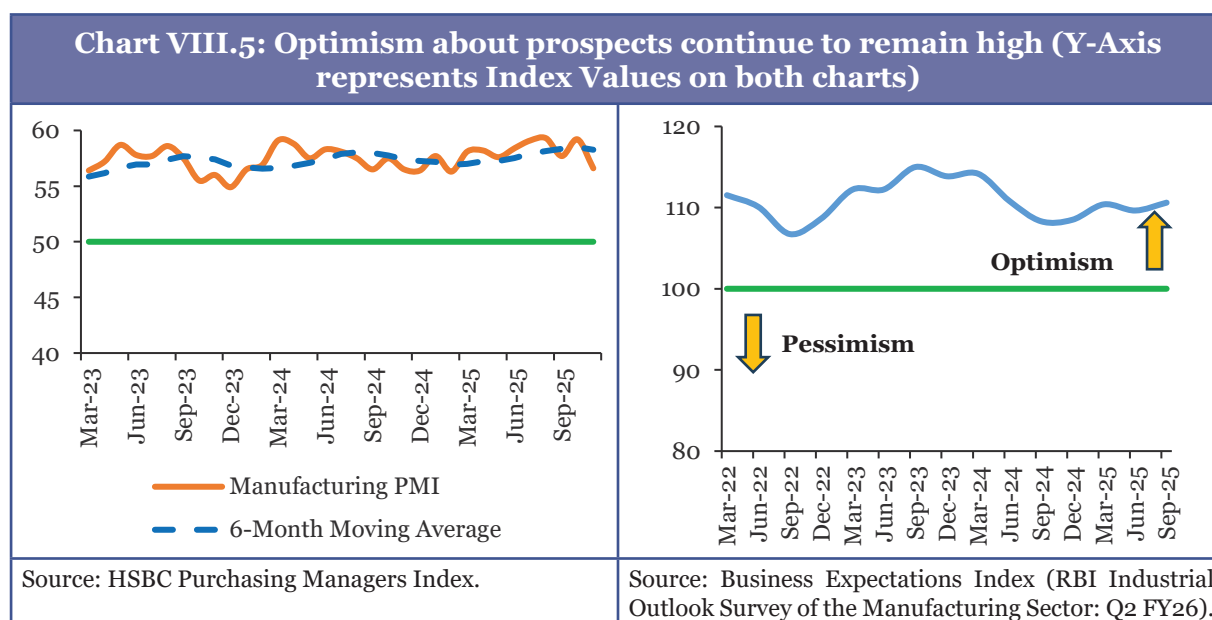
Chart VIII.4: India's Seasonally Adjusted Global Exports of Medium- and High-Tech Manufactured Products (USD Billions)



Source: UNIDO Statistics.

8.10. Forward-looking indicators continue to signal optimism within India's industrial sector. The Manufacturing Purchasing Managers' Index (PMI) has remained well above the expansion threshold of 50 between March 2023 and the present, with both the headline index and its six-month moving average fluctuating comfortably in the mid-50s to high-50s range (Chart VIII.5). This stability suggests broad-based confidence in current operating conditions, order books, and production plans. A similar trend is also seen in the RBI's Business Expectations Index, which has consistently stayed above the neutral benchmark of 100 through FY25 and into Q2 FY26, signaling positive sentiment regarding future output, employment, and investment. Together, these indicators reinforce the continued resilience of industry sentiment and reflect a business environment where firms remain confident about demand and growth prospects.

⁵ UNIDO World Manufacturing Production: Q2 2025 Report.



8.11. Overall, India's industrial sector continues to emerge as a key contributor of resilience and growth amidst the challenging global industrial landscape. Even as the global industrial production grows slowly, India's consistent expansion, growing technological depth, and rising global presence underline its potential to remain a bright spot in the world economy.

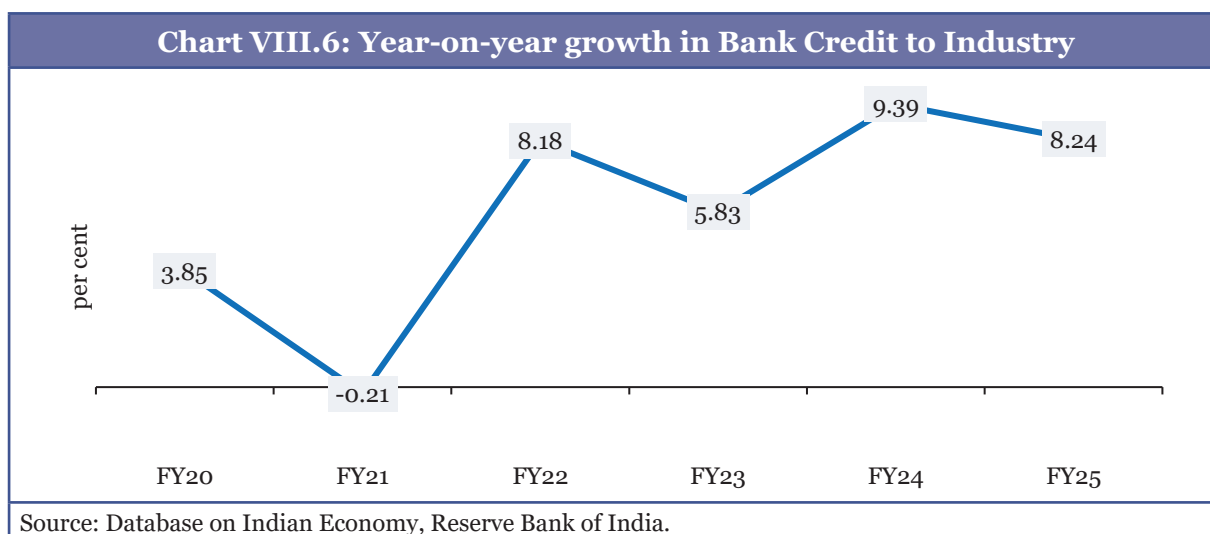
8.12. Given the sector's strategic vitality, significant employment potential, and deep economic linkages, a more concerted focus is required on five structural pillars of competitiveness: strengthening the regulatory environment for ease of doing business; boosting innovation and R&D; addressing skill gaps for future workforce needs; enhancing infrastructure and logistics to lower costs; and scaling up the MSME sector for global competition.

Industrial credit

8.13. Though bank-based industrial credit growth from commercial banks moderated to 8.24 per cent in FY25 compared with 9.39 per cent in FY24, assessments indicate an ongoing diversification of funding sources away from banks. The Monthly Economic Review of August 2025 noted, 'decrease in bank credit coincides with the increase in the overall flow of financial resources to the commercial sector. This shift could be due to the large corporations increasingly relying on market-based instruments such as commercial paper and corporate bonds for their funding requirements'.⁶ Financial flows from non-bank sources to the commercial sector recorded a CAGR of 17.32 per cent over the period FY20 to FY25. Furthermore, the total flow of financial resources from domestic sources to the commercial sector has demonstrated remarkable resilience,

⁶ <https://tinyurl.com/yd2kafsy>.

growing at a CAGR of 28.6 per cent since FY20.⁷ These trends highlight how effectively the financial resources are growing from non-bank sources, offsetting the moderation in credit from bank sources.⁸



8.14. Against this backdrop, the moderation in bank credit flow to industry—though still growing at a rate above 8 per cent in FY25—should be seen as a sign of an ongoing structural adjustment. Diversification of finance sourcing by industries is expected to improve stability, reduce cyclical dependence on bank credit, and align funding more closely with firms’ balance sheet strength. The growing depth of the corporate bond markets and non-bank financing channels is expected to create an industrial credit landscape supportive of higher investment with lower systemic risk.

Table VIII.1: Industry-wise growth in deployment of gross bank credit (in per cent)

Sector	CAGR FY21-FY24	FY25 Growth (y-o-y)
Mining and Quarrying (incl. Coal)	7.62%	4.79%
Food Processing	10.09%	5.10%
Beverage and Tobacco	20.48%	14.06%
Textiles	7.70%	8.29%
Leather and Leather Products	5.44%	3.11%
Wood and Wood Products	16.48%	16.73%
Paper and Paper Products	7.04%	13.83%
Petroleum, Coal Products and Nuclear Fuels	18.68%	16.49%
Chemicals and Chemical Products	10.47%	7.41%
Rubber, Plastic and their Products	17.32%	14.36%

⁷ RBI Handbook of Statistics on Indian Economy (2024-25), Table 63, available at <https://tinyurl.com/2rme64wd>.

⁸ RBI Bulletin, October 2025: <https://tinyurl.com/yjehzk5y>.

Glass and Glassware	22.46%	11.19%
Cement and Cement Products	1.23%	-0.01%
Basic Metal and Metal Product	7.97%	12.76%
All Engineering	8.64%	21.98%
Vehicles, Vehicle Parts and Transport Equipment	9.75%	5.20%
Gems and Jewellery	4.53%	1.01%
Construction	5.92%	12.54%
Other Industries	5.38%	17.38%

Source: Reserve Bank of India (RBI) Database on Indian Economy.

8.15. The subsequent sections of this chapter are organized as follows. The next two sections provide an overview of progress and policy measures across various industrial segments, including core input industries and capital and consumer goods sectors. This is followed by high-growth sector spotlights, offering in-depth analyses of selected sectors—electronics, automotive, and pharmaceuticals—focusing on exports, value-added activities, and technology adoption, as well as flagship initiatives to promote manufacturing. This is followed by a brief discussion on the cross-cutting structural pillars of competitiveness: innovation and R&D, regulatory reforms [including a review of Quality Control Orders (QCOs)], infrastructure and logistics, and scaling up the Micro, Small and Medium Enterprises (MSMEs). The final section concludes the discussion and presents an indicative roadmap towards greater integration with the GVCs.

CORE INPUT INDUSTRIES

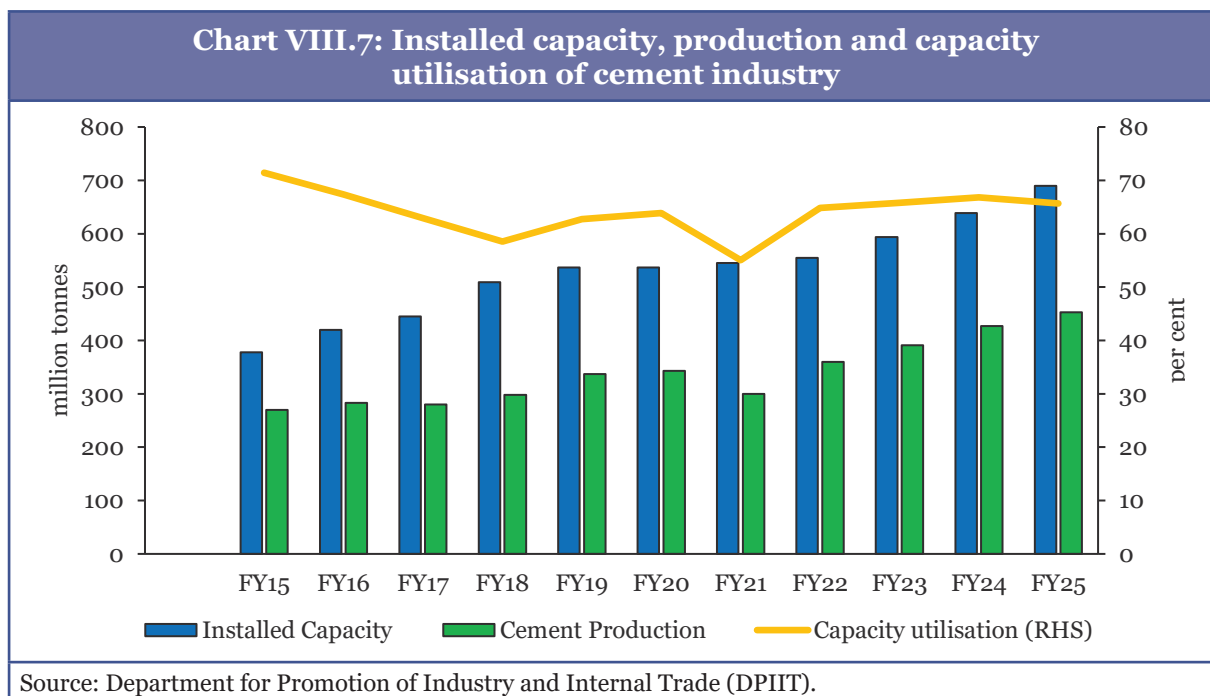
Cement

8.16. India is the second-largest cement producer in the world after China. The Indian cement industry comprises 160 integrated large cement plants, 130 grinding units and 62 mini cement plants. The current annual installed capacity of the cement industry in India is about 690 million tonnes, with cement production of around 453 million tonnes in FY25.⁹ Most of the cement plants in India are located near the raw material sources. About 85 per cent of the cement industry is concentrated in the states of Rajasthan, Andhra Pradesh, Telangana, Karnataka, Madhya Pradesh, Gujarat, Tamil Nadu, Maharashtra, Uttar Pradesh, Chhattisgarh and West Bengal. The industry has adequate installed capacity to meet the domestic demand for cement. Domestic cement consumption in India is approximately 290 kg per capita, compared to a global average of 540 kg per capita.¹⁰

⁹ DPIIT.

¹⁰ *ibid.*

8.17. The cement industry is mainly driven by robust infrastructure development and urbanisation. The Government's focus on mega infrastructure projects such as highways, railways, housing schemes, smart cities and rural development and industrial growth is expected to fuel cement demand significantly.



Steel

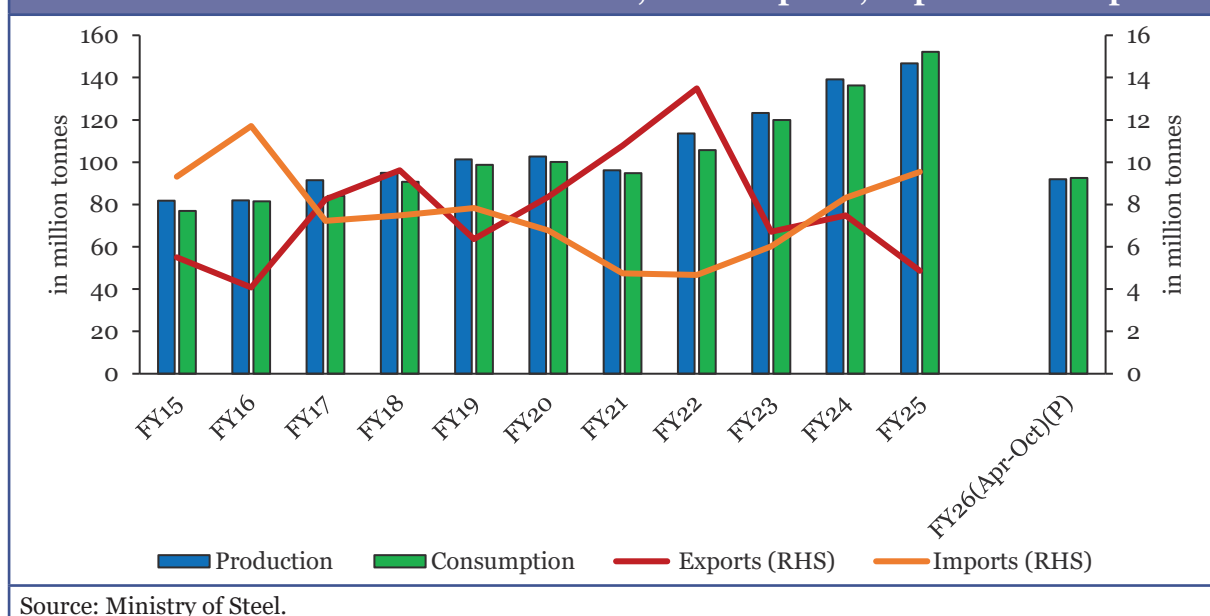
8.18. The steel sector serves as the backbone of industrialisation and infrastructure, securing India's position as the world's second-largest crude steel producer. The sector has undergone a major transformation in the last five years, largely driven by strong domestic demand from the construction and manufacturing sectors.

Table VIII.2: Growth in Steel Production and Consumption

Metric	2020-21 (MT)	2024-25 (MT)	CAGR (per cent)
Crude Steel Production	103.54	152.18	10.1
Finished Steel Production	96.20	146.69	11.1
Steel Consumption	94.89	152.13	12.5
Source: Ministry of Steel.			

8.19. This momentum continued into the fiscal year 2025-26, with crude steel production growing by 11.7 per cent, finished steel production increasing by 10.8 per cent, and consumption rising by 7.8 per cent during April-October 2025-26, compared to the corresponding period last year.¹¹

¹¹ Ministry of Steel.

Chart VIII.8: Finished Steel Production, Consumption, Exports and Imports

8.20. To sustain the sector's robust growth and foster self-reliance, the Government launched the Production-Linked Incentive (PLI) Scheme for Specialty Steel in 2021, with an outlay of ₹6,322 crore, aimed at promoting high-value, niche products. As of October 2025, cumulative investment under the PLI scheme reached ₹23,022 crore, with production of 2.34 million tonnes (MT) of specialty steel.¹²

8.21. However, the sector faces challenges related to international price disparity and raw material security.¹³ India was a net importer of steel during FY26 (April–October), primarily due to low international prices which resulted in lower margins on exports and cheaper imports. While India is largely self-sufficient in iron ore, the industry faces critical dependence on imported Coking Coal. To mitigate global supply risks, the Ministry of Coal launched Mission Coking Coal in 2022 with a view to significantly enhance domestic raw coking coal production to 140 MT by 2030.

Coal

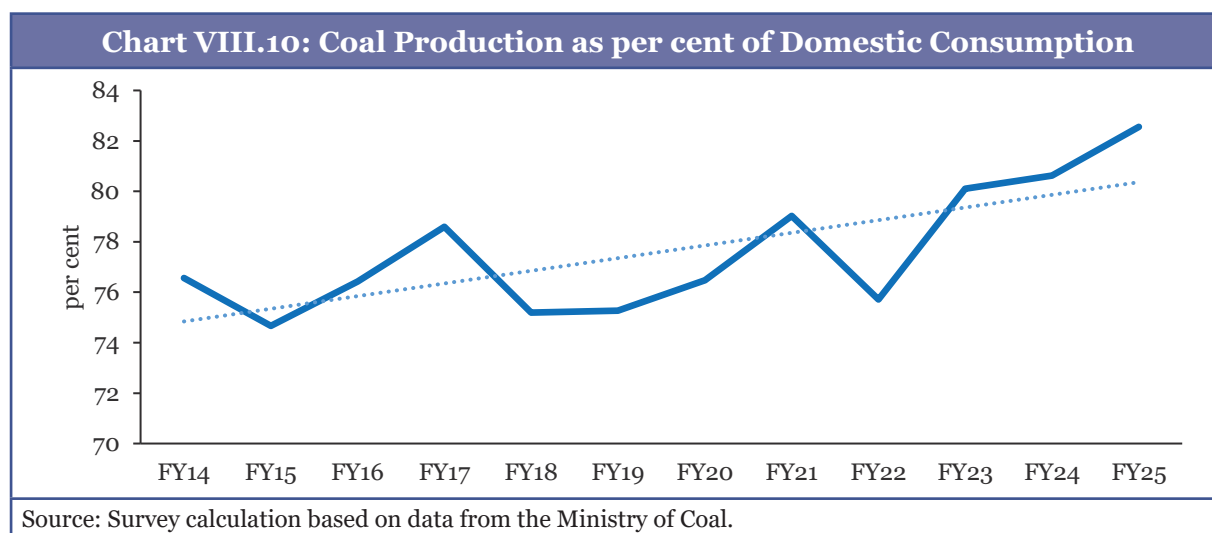
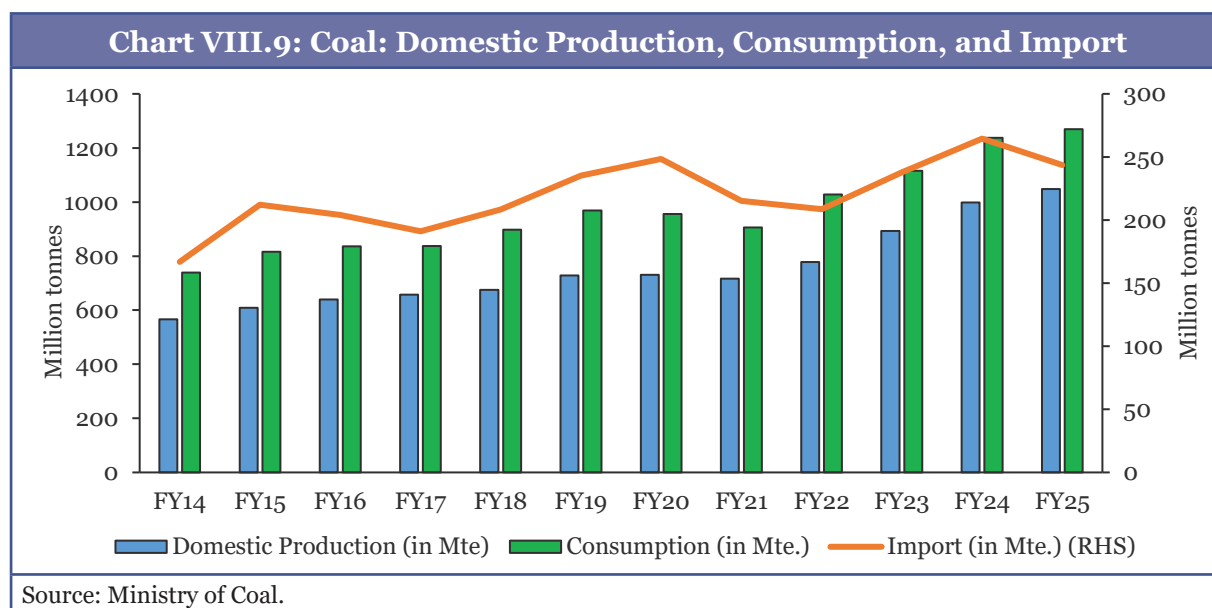
8.22. With the fifth-largest coal reserves and as the second-largest consumer, coal remains crucial, contributing 55 per cent to the national energy mix and fuelling over 74 per cent of total power generation.¹⁴ India's coal industry reached historic heights in FY25, producing 1,047.52 Million Tonnes (MT) of coal—a 4.98 per cent increase from the previous year's 997.83 MT. This represents the highest coal production the country has ever achieved, demonstrating the industry's impressive growth and

¹² *ibid.*

¹³ *ibid.*

¹⁴ <https://tinyurl.com/2ceh389y>.

importance to India's energy needs. Concurrently, the total coal supply surged by 5.38 per cent, reaching 1,025.33 MT in FY25. Driven by this strong domestic output, imports registered a significant 7.9 per cent decline, falling from 264.53 MT in FY24 to 243.62 MT in FY25.¹⁵ Furthermore, the ratio of domestic production to consumption has also steadily improved over the past decade, as production growth has consistently outpaced consumption growth (Chart VIII.10).



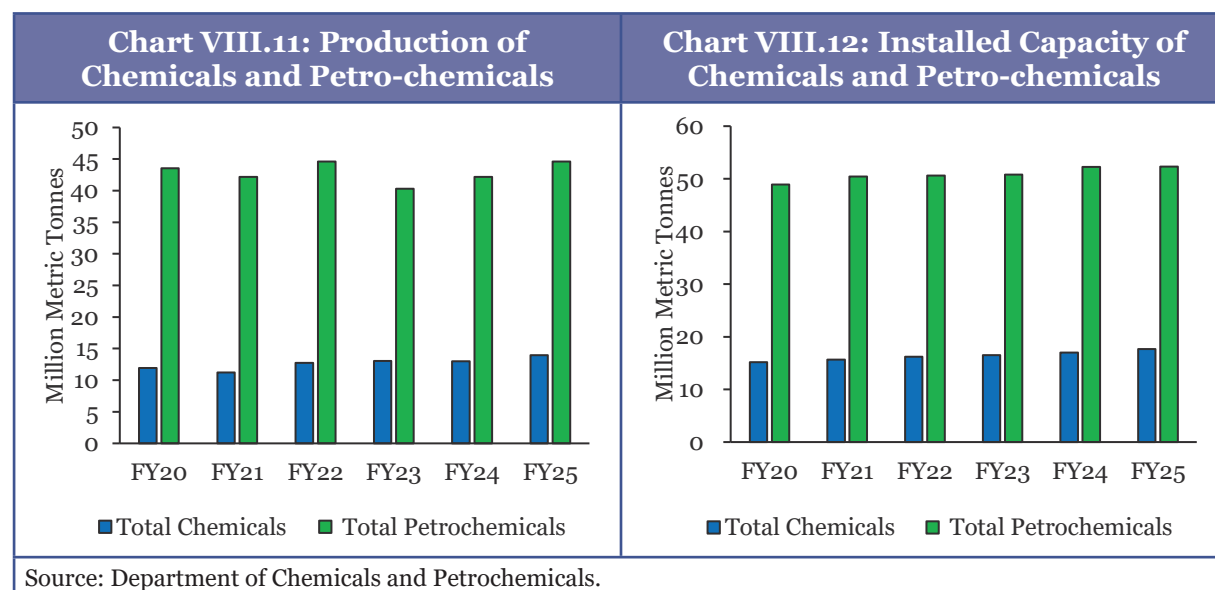
Chemicals & Petrochemicals

8.23. The Chemicals and Petrochemicals sector continues to play a significant role in industrial development of the economy, having strong backward linkages with petroleum refining and natural gas processing, as well as forward linkages with

¹⁵ Ministry of Coal.

numerous downstream industries. The sector contributed 8.1 per cent to the overall manufacturing sector's GVA in FY24.

8.24. The production of selected major chemicals and petrochemicals¹⁶ has demonstrated a consistent upward trend over the last decade. The total production of Major Chemicals and Petrochemicals reached 58,617 thousand MT in FY25, as compared to 45,638 thousand MT in FY16, registering a Compound Annual Growth Rate (CAGR) of 2.8 per cent during the period FY16 to FY25.¹⁷



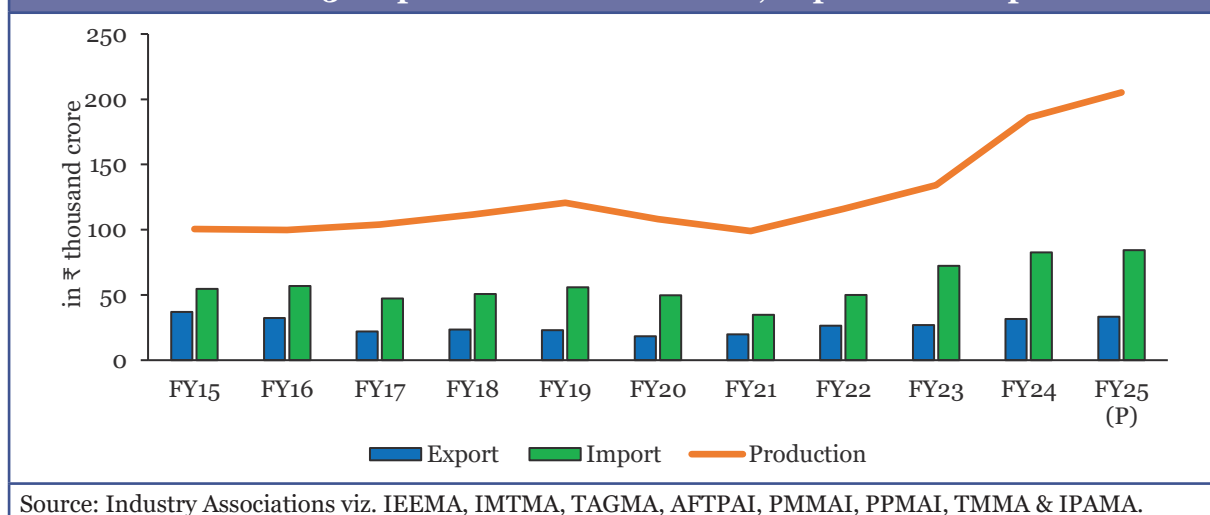
PERFORMANCE OF CAPITAL GOODS AND CONSUMER GOODS INDUSTRIES

Capital Goods

8.25. The export performance of the Capital Goods sector since FY22 is closely aligned with the expansion in domestic production capacity and the strengthening of domestic investment activity. Simultaneously, imports have registered consistent growth, reflecting robust domestic investment demand, as well as a reliance on technologically advanced imported machinery. Overall, these dual trends indicate strong domestic investment cycles but also indicate the continuing need to reduce dependence on imported high-tech capital goods.

¹⁶ Department of Chemicals and Petrochemicals monitors selected Major Chemicals and Petrochemicals products only. The data source in respect of Installed Capacity and Production of Chemical products is Monthly Production Reports (MPRs) received from the chemical manufactures under large and medium scale units only.

¹⁷ Department of Chemicals and Petrochemicals.

Chart VIII.13: Capital Goods: Production, Exports and Imports

8.26. The Government is actively trying to enhance the competitiveness of the Capital Goods Sector through Phase II of the Scheme on Enhancement of Competitiveness in the Indian Capital Goods Sector, launched in 2022. The scheme focuses on six core components, including the identification of technologies through technology innovation portals, the establishment of new Advanced Centres of Excellence (CoEs), Common Engineering Facility Centres (CEFCs), the upgradation of existing testing and certification centres, alongside robust skilling initiatives. Under Phase II of the Scheme, a total of 29 projects with total cost of ₹891.37 crore with government contribution of ₹714.64 crore have been sanctioned, as of November 2025. Many technologies developed under the scheme have also captured export markets in countries such as France, Belgium, Qatar and others.¹⁸

Automobile

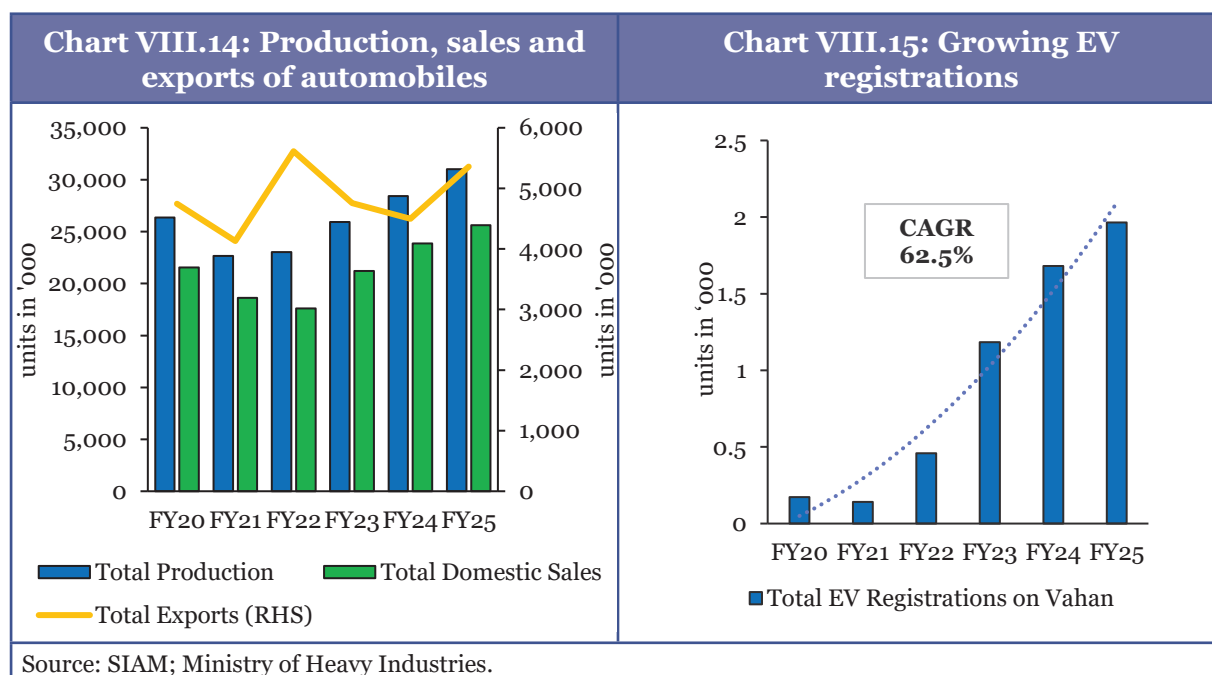
8.27. The automotive industry is a significant driver of economic growth, with India established as the world's largest market for Two-Wheelers and Three-Wheelers and the third-largest market globally for Passenger Vehicles (PV) and Commercial Vehicles (CV). Supported by a vast manufacturing and auto component ecosystem, the sector provides direct and indirect employment to over 30 million people and contributes significantly to public finances, accounting for nearly 15 per cent of the country's GST collections.¹⁹

8.28. The industry is also witnessing tremendous growth in exports, with more than 5.3 million vehicles shipped across Passenger, Commercial, Two-Wheeler, and Three-Wheeler segments in the FY25 and posting double-digit growth in the H1 of 2025-26, reflecting rising global acceptance of India-made vehicles. Overall, the industry has

¹⁸ Ministry of Heavy Industries.

¹⁹ *ibid.*

recorded nearly 33 per cent growth in production over the last decade (from FY15 to FY25). A robust demand-side recovery has driven both production growth and sales in the post-pandemic period. Furthermore, government initiatives highlighted in Box VIII.1 below have driven significant growth in Electric Vehicle (EV) registrations in recent years (Chart VIII.15).



Box VIII.1: Strategic Policy Interventions for Electric Mobility

- **PLI Scheme for Automobile & Auto Components Industry (PLI-Auto Scheme)** approved in September 2021 with ₹25,938 crore outlay incentivises high-value Advanced Automotive Technology (AAT) vehicles and products, and has attracted a cumulative investment of ₹35,657 crore till September 2025.
- **PLI scheme for 'National Programme on Advanced Chemistry Cell (ACC) Battery Storage' (PLI ACC Scheme)** with ₹18,100 Crore outlay for 50 GWh capacity is localising ACC manufacturing, with 40 GWh capacity already awarded, strengthening the EV ecosystem.
- **PM E-DRIVE Scheme:** Launched in September 2024 with an outlay of ₹10,900 crore, which provides demand incentives for e-2W and e-3W, and extends support to new categories, including e-trucks and e-ambulances, alongside funding for charging infrastructure and testing agency upgrades.
- **PM e-Bus Sewa-Payment Security Mechanism (PSM) Scheme** notified in October 2024 with an estimated financial outlay of ₹3,435.33 crore, to support deployment of over 38,000 electric buses. The objective of the Scheme is to provide payment security in case of default by public transport authorities for making timely payment to selected bidders/operators/OEMs on their monthly payment obligations for operation of electric buses.²⁰

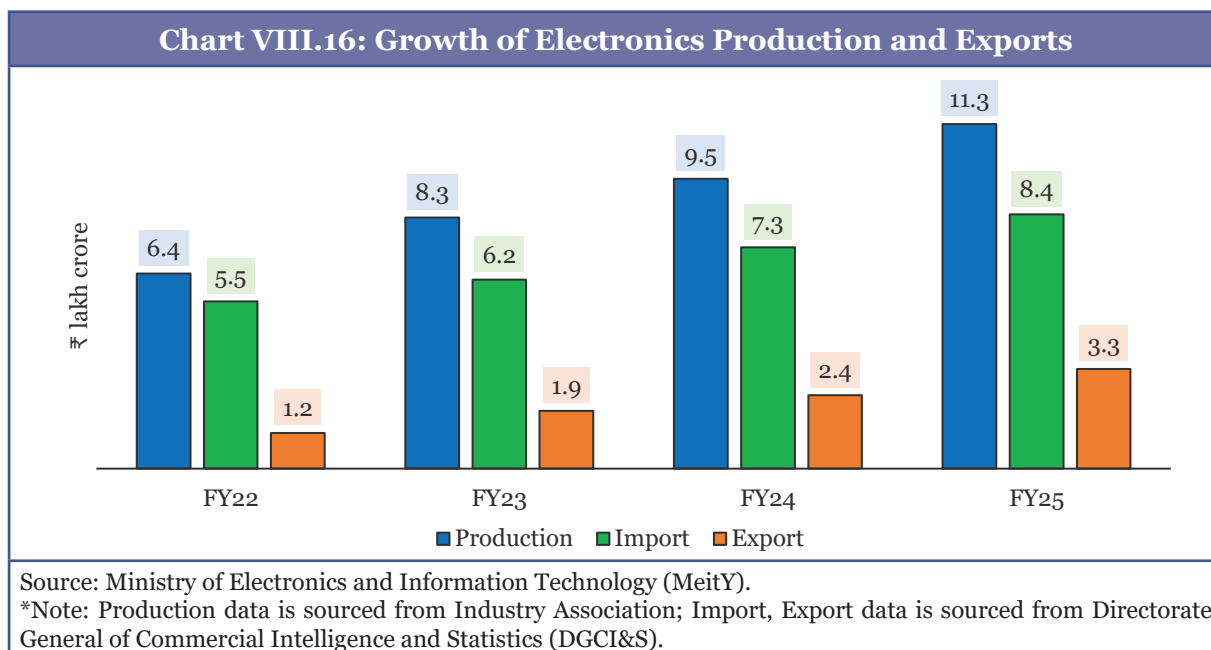
²⁰ <https://tinyurl.com/45j5ye3u>.

- **Scheme to Promote Manufacturing of Electric Passenger Cars in India (SMEC)**, notified in March 2024 aims to attract investments from global EV manufacturers and promote India as a manufacturing destination for e-cars, by allowing temporary imports of high-value e-4W at a reduced customs duty, contingent upon a minimum investment of ₹4,150 crore (USD 500 million) and achieving mandatory phased Domestic Value Addition (DVA) targets.

Electronics

8.29. India's electronics sector has undergone a structural transformation in recent years, ascending from the seventh-largest export category in FY22 to the third-largest and fastest-growing in FY25. This momentum continued into the first half of FY26, with electronics exports reaching USD 22.2 billion, positioning the sector on track to become the second-largest exported item.²¹

8.30. This growth is underpinned by a remarkable surge in domestic production and export volumes (Chart VIII.16). Central to this expansion is the mobile manufacturing segment, which witnessed a nearly 30-fold increase in production value, rising from ₹18,000 crore in FY15 to ₹5.45 lakh crore in FY25. Transitioning from a net importer to the world's second-largest mobile phone manufacturer, India now hosts over 300 manufacturing units, a significant leap from just two units in 2014.²²



8.31. The government has also recognised the challenges inherent in building a competitive electronics manufacturing ecosystem—particularly the need for large

²¹ Source: PIB Release available at <https://tinyurl.com/yy2z5zh5>.

²² *ibid*.

capital investment, economies of scale, long gestation periods, access to cutting-edge technologies, and a highly skilled workforce. In response, a series of strategic interventions have been implemented (Box VIII.2) to enable domestic firms to overcome these barriers and integrate more effectively into GVCs.

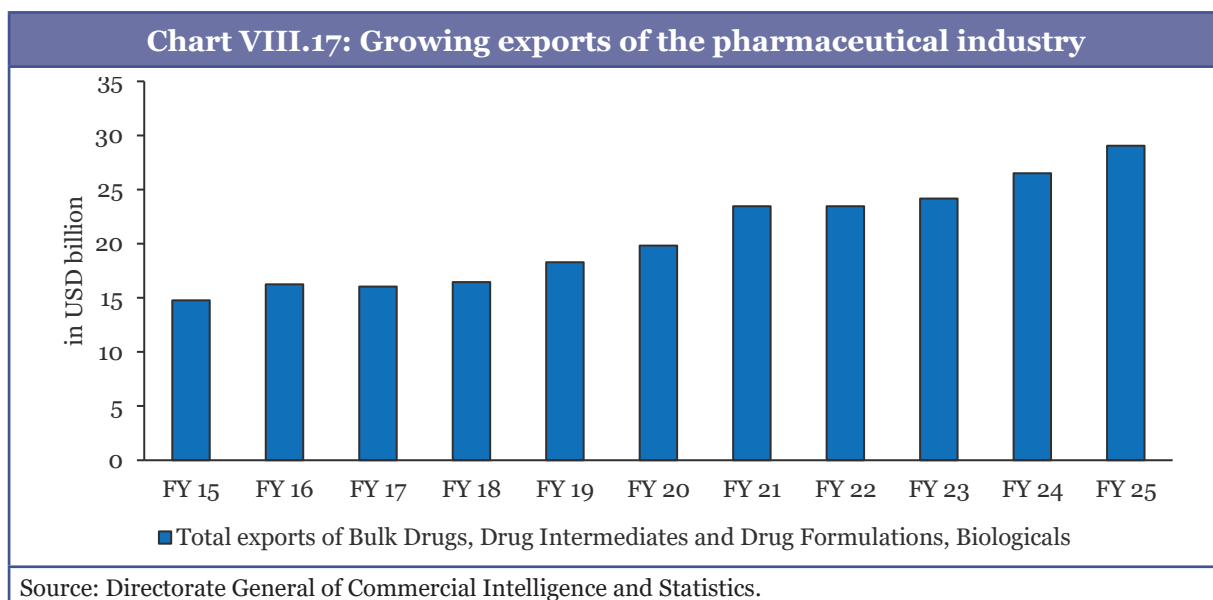
Box VIII.2: Key initiatives to strengthen electronics manufacturing

- **PLI Scheme for Large Scale Electronics Manufacturing:** Launched in April 2020, the scheme has generated a cumulative production of approximately ₹9.34 lakh crore, exports of ₹5.12 lakh crore and investment of ₹13,759 crore, as of September 2025.
- **PLI Scheme 2.0 for IT Hardware:** Launched in May 2023, the PLI schemes for IT Hardware have realised a cumulative production of ₹14,462.7 crore and investments of ₹892.47 crore as of September 2025.
- **Electronics Component Manufacturing Scheme (ECMS):** Notified in April 2025 with a budgetary outlay of ₹22,919 crore, the scheme aims to develop robust component manufacturing ecosystem by integrating domestic electronic industry with GVCs. It offers (a) turnover-linked incentive, (b) capex incentive and hybrid incentive (combination of (a) and (b)) on target segment products.
- **Electronics Manufacturing Clusters (EMC and EMC 2.0) Scheme:** These schemes provide "plug-and-play" infrastructure to attract global firms. Under EMC, 19 Greenfield clusters and 3 Common Facility Centers (CFCs) were approved, while EMC 2.0 has approved 11 clusters and 2 CFCs, as of September 2025.
- **Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors (SPECS):** Notified in April 2020, it provides a 25 per cent financial incentive on capital expenditure for the downstream value chain. As of September 2025, 58 applications have been approved with a total proposed investment of ₹22,081 crore.
- **Modified Programme for Development of Semiconductors and Display Manufacturing Ecosystem:** Under the ₹76,000 crore programme, the India Semiconductor Mission (ISM) has approved landmark projects, including Micron's ATMP (Assembly, Testing, Marking and Packaging) facility, Tata Electronics' Semiconductor Fab, one compound semiconductor fab and various other packaging facilities, as a first step towards high-tech self-reliance. In addition, 24 projects for financial support and 100 companies for design infrastructure support for chip design to be implemented by domestic start-ups/ MSMEs have been approved.

Pharmaceuticals

8.32. The Indian pharmaceutical industry is the world's third-largest by volume, meeting approximately 20 per cent of global generics demand, with exports to 191 countries in FY25. Over 50 per cent of these exports are directed to highly regulated

markets such as the United States and Europe. Beyond generics, India is a global leader in low-cost vaccine supply, providing a majority of the world's diphtheria, tetanus and pertussis (DPT), Bacillus Calmette-Guerin (BCG) and measles vaccines. In FY25, the sector's annual turnover reached ₹4.72 lakh crore, with exports growing at a CAGR of 7 per cent over the last decade (FY15 to FY 25) (Chart VIII.17).²³



8.33. Moreover, India's medical devices sector is also rapidly becoming globally competitive, with exports to 187 countries in FY25. The industry now manufactures high-end equipment, including MRI and CT scanners, linear accelerators, cardiac stents, and ventilators. This expansion into sophisticated imaging and life-support technologies marks a significant shift toward high-tech medical manufacturing. Key initiatives for both sectors are detailed in Box VIII.3.

8.34. India currently ranks 11th globally in pharmaceutical exports by value, with a 3 per cent share and medical devices exports have grown significantly from USD 2.5 billion in FY21 to USD 4.1 billion in FY25, though there exists substantial scope for further expansion.²⁴ To move up the value chain, the pharmaceutical industry is shifting from a volume-driven to a value-driven approach, with greater emphasis on complex generics, biosimilars, and innovation. Likewise, scaling up the medical devices sector necessitates reducing import dependence through the adoption of advanced manufacturing technologies such as AI and 3D printing, along with streamlining global certification processes to strengthen international market access.

²³ Department of Pharmaceuticals.

²⁴ *ibid.*

Box VIII.3: Key policy initiatives in the pharmaceutical sector

- **Supply Chain Resilience (PLI for Bulk Drugs):** To mitigate reliance on imported Active Pharmaceutical Ingredients (APIs) or their Key Starting Materials (KSMs) or Drug Intermediates (DIs), the PLI scheme for Bulk Drugs has mobilised investments worth ₹4,763 crore as of September 2025 and created manufacturing capacity of 55,000 MT per year for 26 critical products, including a strategic focus on fermentation-based synthesis for KSMs like Penicillin G Potassium, which is the precursor for several semi-synthetic antibiotics, such as Amoxicillin, Ampicillin, Cloxacillin and Dicloxacillin.
- **Strengthening of Pharmaceutical Industry (SPI):** With an outlay of ₹500 crore, this scheme supports MSMEs and clusters. Under the Revamped Pharmaceuticals Technology Upgradation Scheme (RPTUAS), 255 applications have been approved to facilitate units align with global standards.
- **Scheme for Promotion of Bulk Drug Parks:** Three world-class parks are being developed in Gujarat, Himachal Pradesh, and Andhra Pradesh with a total ₹3,000 crore outlay.
- **PLI for Medical Devices:** As of September 2025, the scheme has attracted ₹1,093.69 crore in actual investment, and manufacturing of 57 high-end medical devices has started in the country.
- **Scheme for Promotion of Medical Devices Parks:** Final approval for financial assistance of ₹100 crore each has been given to the states of Uttar Pradesh, Tamil Nadu and Madhya Pradesh. Under the scheme, the land has already been allotted to 184 manufacturers.
- **Strengthening of Medical Device Industry (SMDI) Scheme:** Launched in November 2024 with a ₹500 crore outlay, this scheme focuses on reducing import dependence, capacity building, and supporting clinical studies.

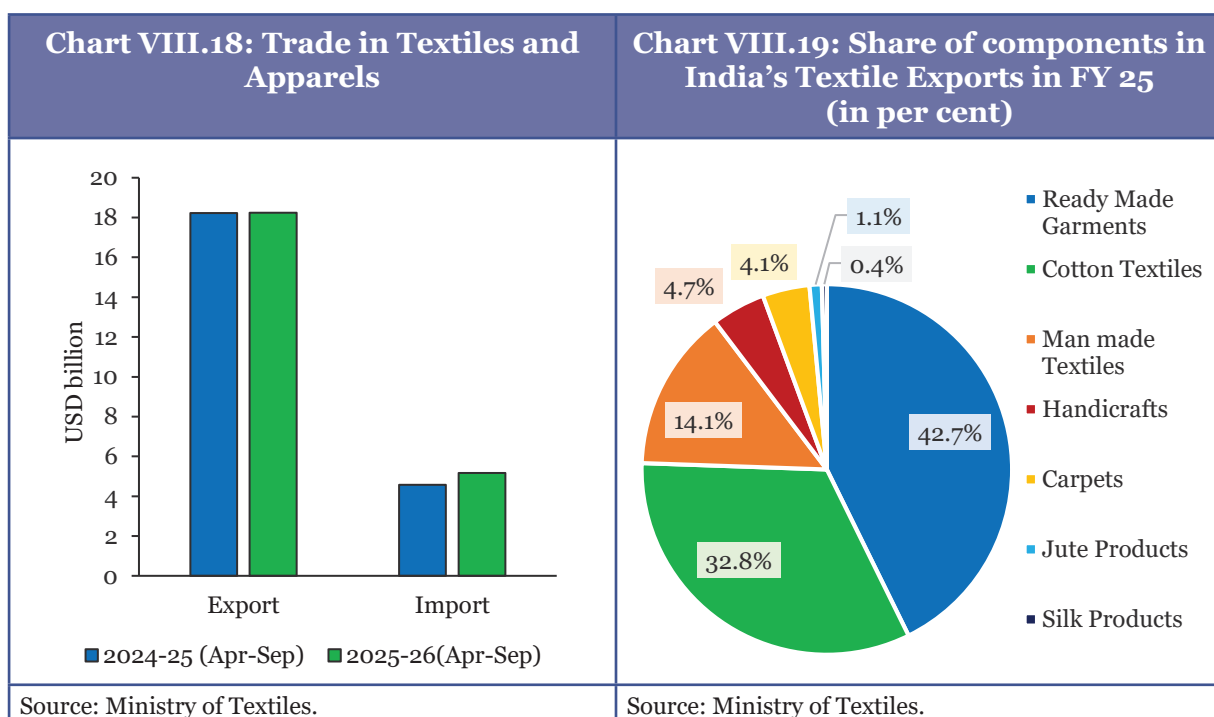
Textiles

8.35. The Indian apparel & textile industry, with a size of about USD 179 billion, contributes close to 2 per cent of GDP and accounts for about 11 per cent of the manufacturing GVA. India is the world's largest cultivator of cotton by acreage and the largest producer of jute, as well as the second-largest producer of silk and cotton. Beyond natural fibres, India is a major global hub in the man-made fibres (MMF) segment, ranking second globally, supported by large, technologically advanced manufacturing facilities. The country is the second-largest producer of polyester and viscose fibres.²⁵

8.36. India is the 6th largest global exporter of textiles and apparel, with a share of about 4 per cent in world exports in this segment. India's export of Textiles and Apparel (including Handicrafts) increased to USD 37.75 billion in FY25, up from USD 35.87 billion in FY24.²⁶

²⁵ Ministry of Textiles.

²⁶ *ibid.*



8.37. While India's textiles sector possesses significant intrinsic strengths, the sector faces challenges such as, small scale and fragmentation of the value chain, heavy reliance on cotton [which contrasts with the global market's focus on Man-Made Fibres (MMF)], limited FDI and technology adoption. To drive a structural transformation, the Government has taken several initiatives such as the launch of the PM-Mega Integrated Textile Region and Apparel Parks (PM-MITRA) Scheme to facilitate economies of scale and reduce logistics costs; revision to the Production Linked Incentive (PLI) Scheme for MMF and Technical Textiles in October 2025 to encourage fresh investments and ease entry barriers such as expansion of eligible MMF and technical textile product lines; reduction in minimum investment thresholds (from ₹300 crore to ₹150 crore and from ₹100 crore to ₹50 crore); and lowering the incremental turnover requirement from 25 per cent to 10 per cent.

8.38. Furthermore, to alleviate raw material constraints and reduce production costs, the Government strategically revoked multiple Quality Control Orders (QCOs) in the MMF and viscose value chains in November 2025.²⁷ Besides, the technical textiles segment has been supported by the National Technical Textile Mission (NTTM), launched with an outlay of ₹1,480 crore (2020-21 to 2025-26), which focuses on R&D, market development, and skill development to position India as a global leader in technical textiles.

²⁷ Department of Chemicals and Petrochemicals (DCPC) revoked 14 QCOs—seven of them relating to the polyester value chain—on 12 November 2025, followed by Ministry of Textiles' revocation of QCO on viscose staple fibre (VSF) on 18 November 2025. [Source: Ministry of Textiles].

KEY INITIATIVES TO PROMOTE MANUFACTURING

Performance Linked Incentive (PLI) scheme

8.39. In alignment with India's vision of 'Aatmanirbhar Bharat', the Production Linked Incentive (PLI) Scheme was launched in 2020 and now spanning 14 key sectors, with an outlay of ₹1.97 lakh crore. The core purpose of these schemes is to attract investments in key sectors and introduce cutting-edge technology, ensure efficiency, and achieve economies of size and scale, thereby making Indian companies and manufacturers globally competitive. Till September 2025, under the PLI scheme, an actual investment of over ₹2.0 lakh crore has been realised, leading to incremental production/sales of over ₹18.70 lakh crore and employment generation of over 12.60 lakhs (direct and indirect). Cumulative incentives of ₹23,946 crore have been disbursed across 12 sectors, with 806 applications approved across all 14 sectors. The PLI Schemes have witnessed exports surpassing ₹8.20 lakh crore, driven significantly by sectors like electronics, pharmaceuticals, and telecom & networking products.²⁸

8.40. Sectoral gains are tangible and worth noting. The electronics sector has emerged as a flagship success story under the PLI strategy. The PLI Scheme has encouraged major smartphone companies to re-locate their production to India. As a result, India has become a major mobile phone manufacturing hub.²⁹ In the first three years, pharmaceutical sales under PLI Scheme for Pharmaceuticals crossed ₹2.63 lakh crore, including exports worth ₹1.69 lakh crore and overall, Domestic Value Addition in the sector has been 83.74 per cent as on March 2025.³⁰ Further, the PLI scheme for automobile and auto-components has attracted cumulative investments worth ₹35,657 crore, resulting in creation of 48,974 jobs, until September 2025.³¹

National Manufacturing Mission

8.41. Complementing PLI, the National Manufacturing Mission (NMM), was announced in Union Budget 2025–26. The NMM aims to boost innovation, raise competitiveness, and expand manufacturing capacity across priority sectors, complementing Make in India and the Atmanirbhar Bharat push. It emphasises faster technology adoption and deeper MSME integration into value chains.

²⁸ Department for Promotion of Industry and Internal Trade (DPIIT).

²⁹ Source: PIB Release available at <https://tinyurl.com/3ffsxy57>.

³⁰ Department of Pharmaceuticals

³¹ Ministry of Heavy Industries.

Box VIII.4: National Mission on Manufacturing (NMM)*³²

The **National Mission on Manufacturing (NMM)**, announced in the Union Budget 2025-26, represents the foundational policy blueprint for accelerating India's industrial growth and global competitiveness over the next decade. It sets ambitious targets for 2035, aiming to double the manufacturing sector's contribution to GDP from 12.9 per cent (2023) to 25 per cent and generate 143 million jobs, along with boosting merchandise exports to USD 1.2 trillion by deepening integration into GVCs.

The Mission has a strategic two-pronged approach, dedicating major focus to sector-specific interventions across 20 to 30 prioritised industrial clusters. This would entail identifying industrial clusters based on parameters such as demand potential, employment generation capacity, and presence of natural endowments and classifying sectors into three archetypes: Scale (e.g., Automotive, Pharmaceuticals) for rapid expansion; Fix and Transform (e.g., Electronics, Capital Goods) for structural reform; and Seed (e.g., Semiconductors, Electric Vehicles) for strategic, innovation-led growth.

The other core activities are expected to address vital, cross-cutting challenges essential for competitiveness, including enhancing the Ease of Doing Business, strengthening Plug & Play Infrastructure, and improving workforce productivity through targeted skilling, empowering MSMEs, prioritising technology and provision of industrial housing. The NMM is thus expected to act as the central facilitator to align efforts across the Centre, States, and clusters, ensuring unified direction and convergence.

Innovation and R&D: From Adopters to Innovators

8.42. India's research and innovation ecosystem has, over the years, strengthened significantly, as its standing in scientific research output has risen sharply from the 7th position in 2010 to the 3rd position globally currently in terms of scholarly publications, reflecting a sustained expansion of academic research capacity.³³ The quality of output is also growing: the number of Indian universities in the 2026 Quacquarelli Symonds (QS) World University Rankings saw a five-fold jump, from 11 institutions in 2015 to 54 in 2026, making India the fourth most-represented country in the rankings.³⁴

8.43. The country's ranking in the Global Innovation Index (GII) improved from 66th in 2019 to 38th in 2025. This places India at the top of the rankings within the lower-middle-income country group and first in the Central and Southern Asia region.³⁵

³² Source: NITI Aayog. *The National Mission on Manufacturing (NMM) was announced in the Union Budget 2025-26. All objectives, components, and institutional arrangements described herein are reported to be at the proposed level.

³³ Department of Science and Technology.

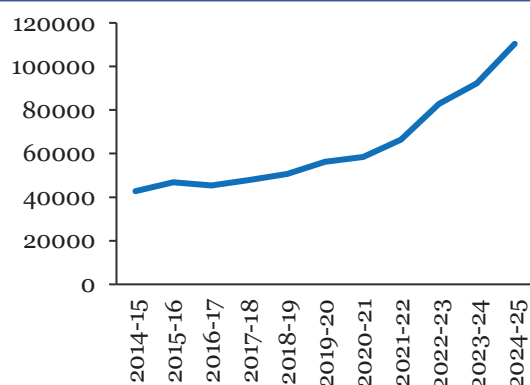
³⁴ PIB Release available at <https://tinyurl.com/3m44bb4m>.

³⁵ Global Innovation Index 2025. World Intellectual Property Organization. <https://tinyurl.com/3ut3cn58>.

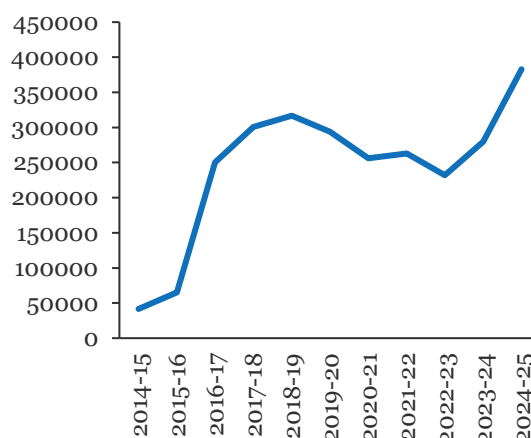
Chart VIII.20: India's Ranking in the Global Innovation Index over the years

Source: World Intellectual Property Organization (WIPO).

8.44. Bengaluru, Delhi and Mumbai feature among the top 50 most innovation-intensive clusters in the world³⁶ and the country's innovation output as a whole, has recorded a marked growth. Emerging as a significant global player in Intellectual Property (IP), India ranks 4th in trademarks, 6th in patents, and 7th in industrial designs in global filings in 2024.³⁷ From FY20 to FY25, patent applications filed nearly doubled, and trademark registrations grew 1.5 times. Designs registered increased by 2.5 times. The increased diffusion of design registrations alongside patent activity suggests a critical shift towards differentiated products, process innovation, and branding, marking an economy moving up the value chain.

Chart VIII.21: Patent Applications

Source: DPIIT.

Chart VIII.22: Trademark Registrations

Source: DPIIT.

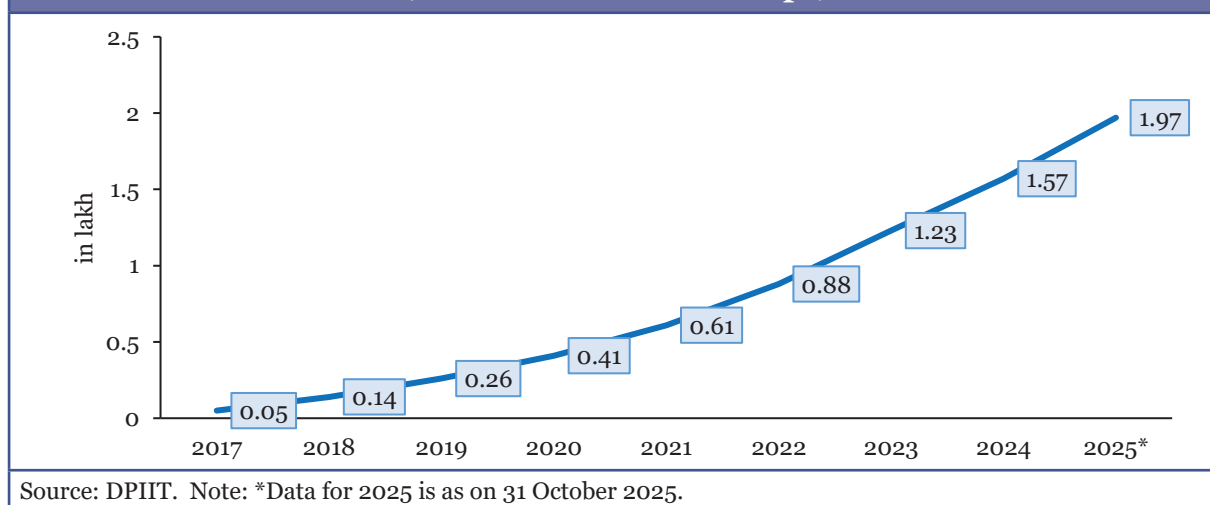
8.45. Marking a shift from the trends of a few years ago, where public-sector drove innovation, R&D now is also being driven by a vibrant start-up ecosystem in the country. The World Intellectual Property Organization (WIPO) ranks India 12th globally, for its

³⁶ *ibid.*³⁷ Source: World IP Indicators 2025, WIPO available at <https://tinyurl.com/bdde84ab>.

entrepreneurship policies and entrepreneurship culture.³⁸ As an evidence to this, since the launch of the Startup India Initiative in 2016, the number of DPIIT-recognised startups has increased from approximately 500 to over 2 lakh as of 2025.³⁹ Flagship financing instruments, such as the Fund of Funds for Startups, the Startup India Seed Fund Scheme, and the Credit Guarantee Scheme for Startups, create an ecosystem that enables entrepreneurs to take risks and innovate with confidence.

8.46. Innovation activity is increasingly broad-based, spanning biotech, Artificial Intelligence (AI), digital services, and sustainability-oriented solutions, supported by a growing pipeline of risk capital, credit access, and incubator-led early-stage support. Collectively, these indicators signal a system that is expanding in scale, quality, and sophistication, with innovation activity graduating from niche research entities into a wider industrial base.

**Chart VIII.23: Rapid growth in DPIIT-Recognised Startups
(Cumulative no. of Startups)**



Box VIII.5: CSIR Progress in Applied Innovation

Over the last year and a half, between FY25 and the first half of FY26, the Council of Scientific and Industrial Research (CSIR) has demonstrated tangible progress in advancing applied research and translating it into deployable solutions across strategic sectors including infrastructure, health, chemicals, energy, aerospace, biotechnology and defence. Shifting from laboratory-oriented research towards impact-oriented research and facilitating technology transfer, CSIR has been working steadily towards supporting national goals of Aatmanirbharta and technology-led growth.

A key driver of CSIR's recent progress has been mission-mode innovation in infrastructure and materials. Indigenous technologies such as REJUPAVE and the ECOFIX pothole repair

³⁸ Economy Profile: India, WIPO: <https://tinyurl.com/29h8jd72>.

³⁹ Inputs received from DPIIT and Startup India Dashboard. As of 25th November, 2025.

system, designed for challenging terrain, are already deployed by the Border Roads Organisation (BRO) in high-altitude regions and are being increasingly adopted by state agencies. Complementing these are early deployments of India's first bio-bitumen highway, steel-slag road infrastructure, and trials with end-of-life plastics, all signalling a clear shift towards circular economy-aligned public works and context-specific applied innovation.

CSIR's applied innovation portfolio now extends into industrial processes, strategic materials, and health technologies with clear deployment pathways. The continuous-zero flow, zero-liquid discharge paracetamol production technology developed by CSIR-National Chemical Laboratory has already been licensed for commercial use, reducing manufacturing costs by an estimated 15-20 per cent. In defence and chemicals, CSIR-Indian Institute of Chemical Technology's (IICT) development of an indigenous process for CL-20 propellant materials strengthens self-reliance in high-energy materials. In biotechnology and health, CSIR's innovations range from an affordable molecular diagnostic for sickle cell anaemia to advances in gene editing platforms and targeted drug delivery for cancer. Collectively, these developments reflect a clear shift from laboratory research to deployable and commercially relevant technologies that support nation-building.

CSIR's work in applied innovation also extends to frontier technologies. The successful test flights of solar-powered high-altitude pseudo satellites (HAPS) capable of long-endurance flight for border security and telecom relay demonstrate deep capability development in autonomous systems. Likewise, indigenous developments in UAV propulsion engines, kamikaze drones, and the partnership with Bharat Electronics Limited for sensor systems demonstrate increasing linkages between public research and national defence capability. CSIR-NAL's technology transfer for commercial production of indigenously designed trainer aircraft (HANSA-3 NG) is another milestone, marking the first time a homegrown civil aircraft platform is moving to commercial manufacturing.

CSIR's recent progress demonstrates a maturing model of government-led applied innovation, where public R&D is increasingly moving beyond publication and prototyping toward deployment, industrial adoption and commercial transfer. By facilitating the transition of technologies from research labs to real-world deployment, this approach enhances technological self-reliance through faster innovation diffusion while broadening India's capacity in strategic and future-oriented fields.

8.47. India is also establishing itself as a strong contender in critical technologies. The Critical Technology Tracker, published by the Australian Strategic Policy Institute (ASPI)⁴⁰, shows India now ranks among the top five countries in 45 out of 64 critical technologies, compared to just 4 technologies in the period from 2003 to 2007. This breadth across defence, space, quantum computing, AI, and advanced materials signals an expanding strategic capability essential for Aatmanirbharta.

⁴⁰ <https://tinyurl.com/4fkhmjc9>.

Box VIII.6: India's Rankings* in Critical Technology Research Output

Defence, space, robotics & transportation	• 3rd in Advanced Aircraft Engines • 3rd in Autonomous underwater vehicles • 4th in Drones, Swarming and collaborative robots
Quantum Computing	• 3rd in post-quantum cryptography • 4th in Quantum Sensors
Biotech, gene tech and vaccines	• 2nd in Biological Manufacturing • 3rd in Novel Antibiotics and Antivirals
Artificial Intelligence, Computing and Communications	• 2nd in Mesh and Infrastructure Independent Networks • 3rd in Advanced radiofrequency communication, Protective cybersecurity technologies • 3rd in AI algorithms, machine learning, Natural Language Processing
Advanced Materials	• 2nd in Advanced Composite Materials, Smart Materials • 3rd in Nanoscale Materials, Coatings, Advanced Explosives
Energy	• 2nd in Biofuels • 3rd in Supercapacitors, Nuclear waste management & recycling • 4th in Nuclear Energy, Hydrogen and ammonia for power

*Note: ASPI's rankings are research-output based and indicate relative scientific leadership and do not necessarily capture industrial maturity, commercial deployment, or production capabilities.
Source: ASPI Critical Technology Tracker.

8.48. Despite notable progress in research and development, challenges remain, primarily concerning R&D expenditure intensity. India's Gross Expenditure on R&D (GERD) as a percentage of GDP stands at a modest 0.64 per cent, substantially below the global average. Leading economies, such as the US (3.48 per cent), China (2.43 per cent), and South Korea (4.91 per cent), invest significantly more.

8.49. Low expenditure in R&D is partly due to low investment in R&D from business sector, which accounts for only 41 per cent of the total expenditure. This is in stark contrast to countries such as China (77 per cent), United States (75 per cent), and South Korea (79 per cent), where business sector contributions to R&D are significantly

higher.⁴¹ Bridging this disparity through various measures and fostering a conducive environment for private industry is critical for accelerating technological development. To drive this endeavour, the government has instituted a series of high-stakes, mission-driven initiatives.

8.50. A major institutional reform driving India's R&D and aimed at addressing the challenges highlighted above is the establishment of the Anusandhan National Research Foundation (ANRF) under the ANRF Act, 2023. The ANRF is intended to provide strategic direction, competitive funding opportunities and collaboration pathways across industry, academia and government. The initiatives taken up by ANRF are being complemented by a suite of mission-driven national programmes such as the National Quantum Mission, the National Mission on Interdisciplinary Cyber-Physical Systems, the IndiaAI Mission, the India Semiconductor Mission, and the National Green Hydrogen Mission. Each initiative focuses on building foundational scientific capability in sunrise domains, translating research into scalable industrial capacity.

8.51. To finance innovation at scale, the Government also announced a new Research, Development and Innovation (RDI) Fund with a total outlay of ₹1 lakh crore over six years and ₹20,000 crore allocated for FY26. The Fund is designed to catalyse private investment in high-tech R&D, support projects at advanced technological readiness levels, enable acquisition of strategically important technologies, and operationalise a Deep-Tech Fund of Funds. Together, the ANRF, the national missions and the RDI Fund form a consolidated architecture for expanding India's GERD beyond its current level and accelerating India's transition towards technological leadership and the broader objectives of Atmanirbhar Bharat.

Box VIII.7: Why the India Semiconductor Mission Matters

Microprocessors are products that are largely obscured from view in our day to day lives. Yet, they form the heart of our energy systems, financial markets, telecom networks, manufacturing units, hospitals, transportation networks, and satellites, among many other applications. The COVID-19 pandemic exposed the vulnerabilities in global semiconductor supply chains. Shortages impacted over 169 industries globally, causing price increases and production delays.⁴² The absence of a steady supply of semiconductors is likely to hamper economic activities across sectors. Reliance on a handful of suppliers for semiconductors is an added risk. The India Semiconductor Mission (ISM), launched in 2021 aims to build domestic capacity across design, manufacturing, and innovation in semiconductors and is a first step towards self-sufficiency and technological sovereignty.

⁴¹ Department of Science and Technology (DST).

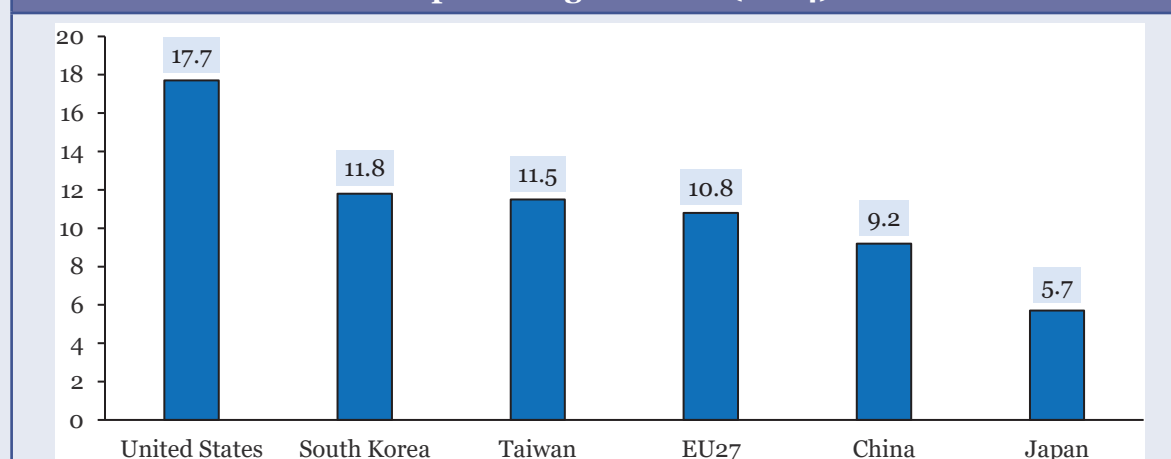
⁴² Economic Survey 2021-22.

Design

Semiconductor manufacturing begins with the design process, which determines the chip's architecture, which optimises parameters such as cost, power consumption and processing capabilities, based on the needs of the chip in question, making their designs incredibly complex. Progress in chip design requires consistent mix of multidisciplinary research, substantial investments, years of R&D for each new chip developed, and a highly skilled workforce. This makes semiconductors the most R&D intensive product to produce, even more than pharmaceuticals and software.

The mix of these factors is not something everyone could afford, nor get right, due to which the design process of cutting-edge chips is highly concentrated in a few developed nations. Global leaders in the design process include the United States, South Korea, Taiwan and Japan, who collectively account for 79.4 per cent of the global semiconductor Integrated Circuit (IC) design revenue.⁴³ Their edge is also maintained by the sustained investments in R&D, allowing them to capitalise on existing intellectual property to further their leadership.⁴⁴

Chart VIII.24: R&D Expenditures in semiconductor industry as a percentage of Sales (2024)



Source: State of the U.S. Semiconductor Industry 2025, Semiconductor Industry Association.

Manufacturing of semiconductors

The manufacturing phase of semiconductors is resource-intensive, requiring highly specialised machinery and facilities. Making semiconductors is extremely resource-intensive and depends on very specialised equipment and facilities. A modern semiconductor factory can cost around USD 10 billion to set up, largely because the machines used must work with nanometre-level precision. The supply of the machines capable of undertaking these complex tasks is concentrated with just 5 companies across 4 countries.⁴⁵

43 Mapping the Semiconductor Supply Chain, Akhil Thadani & Gregory C. Allen. Center for Strategic and International Studies. May 2023.

44 Studies have highlighted the case for a more proactive and strategically calibrated engagement with global intellectual property rights (IPR) treaties to support innovation and R&D-led growth (See: Sanjeev Sanyal and Apurv Kumar Mishra, India and Global IPR Treaties, Working Paper, Economic Advisory Council to the Prime Minister, 2024).

45 Semiconductor Lithography Equipment Market Size & Share Analysis – Growth Trends and Forecasts (2024-2029), Mordor Intelligence.

Furthermore, at these nanometre scales, efficient production requires a high degree of control over the entire process, with specialised and sensitive equipment. Even here, the supply chain is highly concentrated. As of 2021, the United States, Japan and South Korea make up two-thirds of the global sales for wafer fabrication, assembly and test equipment.

Recent geopolitical events underscore a broader lesson: economies that depend on external access to foundational technologies increasingly face constraints not just in production, but in strategic autonomy. As advanced economies accelerate investments in semiconductor capacity and tighten export controls, access to chips becomes a determinant of competitiveness and also national security.

The India Semiconductor Mission for a Resilient India

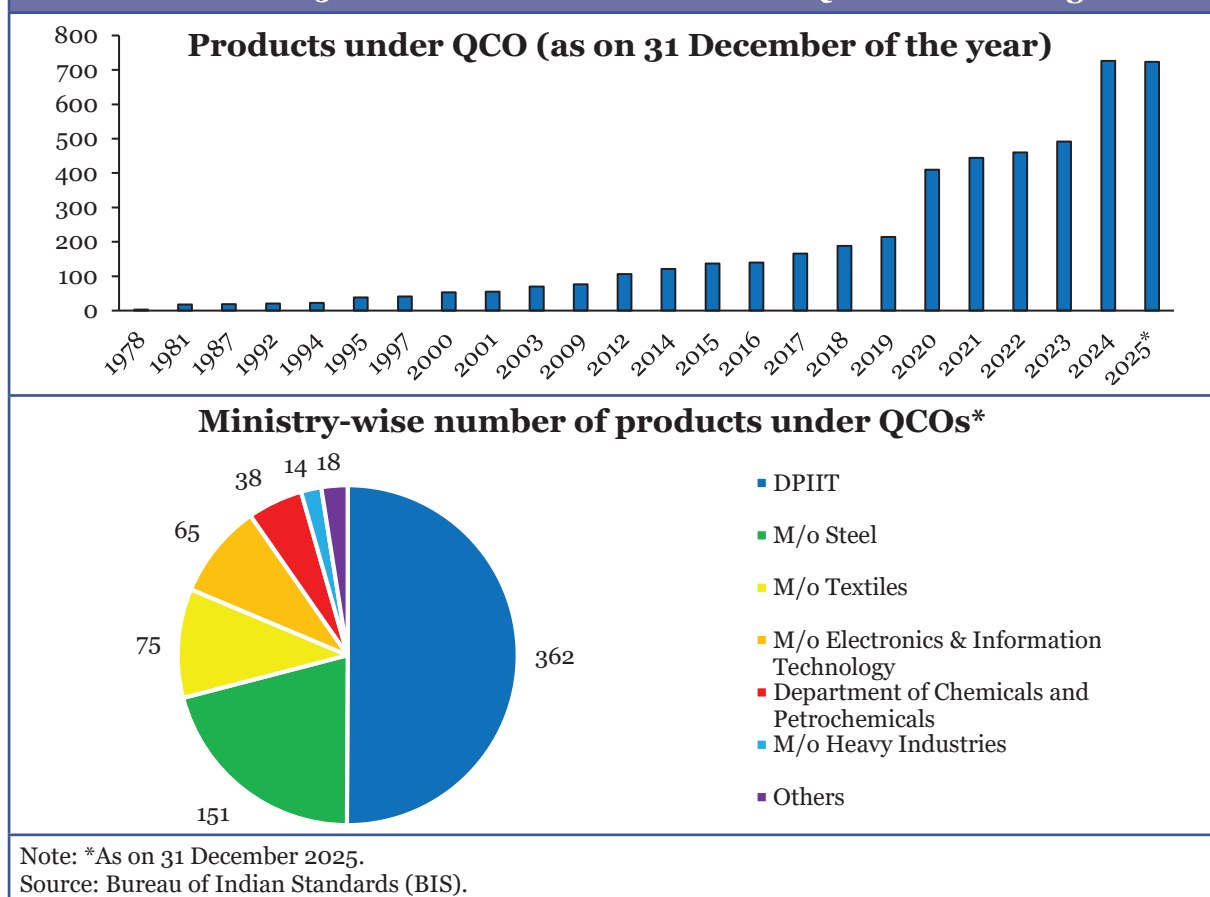
To address structural vulnerabilities and build resilience, the Government of India has undertaken a series of coordinated measures to develop an end-to-end domestic semiconductor system. The India Semiconductor Mission and the Semicon India programme together form the core of this strategy, supported by a ₹76,000 crore incentive framework to catalyse investment in fabrication, assembly, testing, marking and packaging facilities. Under these programmes, four targeted schemes have been launched, each extending fiscal support of 50 per cent of project cost/capex for semiconductor and display fabs, compound semiconductor facilities, and outsourced semiconductor assembly and testing units, alongside a dedicated Design Linked Incentive scheme to encourage domestic chip design. As of August 2025, ten semiconductor manufacturing and packaging projects have been approved with a cumulative investment of around ₹1.60 lakh crore in 6 states.⁴⁶ In addition, State Governments have also begun complementing the national framework, with initiatives such as Odisha's Semiconductor Manufacturing and Fabless Policy offering additional incentives and institutional support. As execution accelerates and capacity expands, these measures are expected to strengthen domestic semiconductor capabilities, enhance the resilience of critical supply chains and position India as a significant participant in the global semiconductor marketplace.

Quality control: Enhancing global competitiveness

8.52. With the goals of the National Manufacturing Mission seeking to position India as an indispensable part of GVCs, quality aspect is as critical as cost competitiveness. In this context, Quality Control Orders (QCOs) are regulatory instruments that mandate conformity with the desired quality standards. The Government of India has significantly augmented mandatory quality assurance in India. As of 31 December 2025, a total of 143 QCOs covering 723 products have been notified by various ministries,⁴⁷ more than tripling the coverage from 214 products in 2019. The purpose of QCOs is to ensure that products meet stipulated quality standards, while avoiding market distortions and protecting consumer requirements.

⁴⁶ <https://tinyurl.com/4hap8fdy>.

⁴⁷ Apart from these, 2 Horizontal QCOs on Safety of all Household Electrical Appliances (as per IS 302 (Part 1)) and Safety of Machinery and Electrical Equipment have also been notified. (Source: Bureau of Indian Standards (BIS))

Chart VIII.25: Growth in terms of number of QCOs and coverage**Box VIII.8: Quality Control-Towards Pragmatism**

In a global environment marked by tightening technical regulations, QCOs are vital for addressing and mitigating the reputational risks associated with inadequate and inconsistent quality. They level the playing field for firms that invest in quality while curbing the influx of substandard imports. In several sectors, the introduction of QCOs for finished goods has been accompanied by improved compliance and a reduction in the circulation of substandard products including sectors such as toys, transformers, footwear, and cement, directly contributing to India's long-term goal of overall improved standards.

A prominent example of this strategic approach leading to a fundamental trade shift is seen in the Toys sector. A Quality Control Order (QCO) for Toys was issued in 2020 w.e.f 01.01.2021, along with an increase in Basic Customs Duty (from 20 per cent to 60 per cent in Feb 2020, and subsequently to 70 per cent in March 2023) and mandatory sample testing of each import consignment to curb the import of sub-standard toys. The results demonstrate a clear structural transformation: between FY15 and FY23, imports of toys declined by 52 per cent, while exports soared by 239 per cent.⁴⁸ Crucially, India transitioned from being a net importer (in FY15) to a net exporter of toys from FY21 onwards. This shift was largely attributed to the quality regulations and tariff policies allowing domestic manufacturers to scale up and meet international quality benchmarks.

⁴⁸ <https://tinyurl.com/4x9jusp7>.

Beyond quality, consumer and workplace safety also provide an equally compelling justification for mandatory standards. QCOs on products ranging from electrical appliances to helmets are essential for protecting public welfare and maintaining market confidence. This regulatory enforcement is to be further supported by enhanced surveillance, expanded testing infrastructure, and digitised certification processes.

While the rationale is clear, the design and implementation of QCOs should be grounded in economic practicality. Mandatory certification, while necessary, is not cost-free and can create friction if implemented without adequate preparation and due diligence, especially given that Micro, Small, and Medium Enterprises (MSMEs) form a significant part of India's industrial base. MSMEs often lack the capital to establish in-house testing facilities or adapt quickly to new certification requirements. Overly narrow transition periods can also lead to production delays, supply disruptions, and rising inventory costs, unintentionally burdening the firms QCOs are meant to support.

This becomes particularly relevant when QCOs extend to raw materials and intermediate inputs. Modern manufacturing is intrinsically linked to global suppliers. Mandating certification for specialised raw materials, intermediate inputs, or spare parts—particularly when domestic alternatives are scarce or non-existent—can inadvertently be counterproductive, causing delays in critical downstream sectors such as automotive, power, and electronics. QCOs on intermediates, therefore, require exceptional caution, with flexibility built in when domestic alternatives do not exist especially in view of complexity of modern global supply chains.

Therefore, QCOs should be approached pragmatically as instruments for strengthening quality assurance, consumer safety and strategic resilience within increasingly complex value chains.⁴⁹ If regulations are applied without comprehensive value-chain analysis, it can trigger cascading consequences, increasing costs and erode cost competitiveness. This interconnectedness was famously illustrated by economist Milton Friedman in his 1980 lecture. Holding up a pencil to the audience, he stated, *"...There is not a single person in the world who could make this pencil... When you go down to the store and buy this pencil, you are in effect trading a few minutes of your time for a few seconds of the time of all those thousands of people."*⁵⁰

A forward-looking QCO framework should, therefore, incorporate a rigorous pre-notification assessment, calibrated transition periods (especially for MSMEs), adequate national testing capacity, and a clear alignment with industry readiness. Critically, where sufficient domestic production capacity does not exist, the framework should provide exemptions or alternative pathways for specialised inputs such as raw materials, intermediates, spare parts, and quantities required for R&D. The Government recognises the importance of a balanced and adaptive approach, with recent regulatory decisions reflecting a willingness to recalibrate QCOs.⁵¹ This balanced approach will allow QCOs to strengthen India's manufacturing capabilities without imposing undue burdens, ensuring that quality regulation remains a strategic enabler of long-term economic growth.

49 World Development Report 2025: Standards for Development (Spotlight 1).

50 Source: Free to Choose (1980) documentary/book, based on Leonard E. Read's 1958 essay "I, Pencil".

51 As per BIS, a total of 46 QCOs covering 51 products have been revoked by concerned regulators in 2025–26, while 3 QCOs covering 63 products and 20 categories of machines and electrical equipment under the Omnibus Technical Regulation (OTR), were deferred or suspended by concerned regulators in 2025–26.

Infrastructure and Logistics: Improving Competitiveness

8.53. A competitive manufacturing ecosystem relies not only on firm-level productivity but also on the efficiency, connectivity and predictability of the infrastructure surrounding it. Over the past few years, the infrastructure push has shifted from project-level execution to systems-level planning. PM GatiShakti represents this structural shift by enabling multi-modal planning through a unified geospatial platform. As of November 2025, 57 Ministries and Departments have been onboarded, and over 1,700 data layers have been integrated into the National Master Plan for allowing planners to identify bottlenecks, optimise routes, and avoid duplication in capital expenditure.⁵² The opening of PM GatiShakti Public and the Unified Geospatial Interface now provides private users with access to 230 curated datasets for planning investments and logistics strategies. This marks a transition away from Government-only planning to a systematic approach of shared national infrastructure intelligence, allowing efficient planning and resource utilisation.

8.54. States and districts have also been adopting this framework at scale, with 27 States notifying State Logistics Policies and 28 Aspirational Districts using the GatiShakti District Master Plan Module for area planning. This will then be extended to all 112 Aspirational Districts in the near term. Complementing this is the National Logistics Policy (NLP) and tools such as Logistics Ease Across Different States (LEADS) and the Unified Logistics Interface Platform (ULIP), which benchmark logistics competitiveness and integrate data across ministries. ULIP currently connects 44 systems across 11 ministries through over 2,000 data fields, with more than 1,700 companies registered and 200 crore API transactions executed. These investments in planning capability, digital integration and logistics intelligence are beginning to improve operational predictability for industry.

8.55. Infrastructure execution is advancing rapidly through the National Industrial Corridor Development Programme, which enables the development of new industrial cities with plug-and-play facilities and logistics hubs focused on multimodal connectivity. Phase-I cities such as Dholera, Shendra-Bidkin, Greater Noida and Vikram Udyogpuri are now operational, with 350 industrial plots allotted and investment of ₹2.02 lakh crore across sectors, including electronics, renewables, pharmaceuticals and EVs.⁵³ Complementing this are sector-specific interventions under the National Logistics Policy, such as the Sectoral Plan for Efficient Logistics for the cement industry. The industry is expected to experience reductions in its logistics costs through efficiency gains from multimodal connectivity and digital transparency.

⁵² Department for Promotion of Industry and Internal Trade (DPIIT).

⁵³ *ibid.*

8.56. India's logistics strategy has transitioned beyond mere asset creation to an integrated ecosystem approach, combining physical infrastructure, digital public platforms (like ULIP), regulatory simplification, and corridor-based industrialisation (PM GatiShakti). This holistic approach aims to minimise logistics friction, financing delays, and coordination losses that previously constrained industrial competitiveness.

8.57. A new joint study by DPIIT and NCAER⁵⁴ estimates logistics costs at 7.97 per cent of GDP for FY24, which is lower than 8.84 and 8.79 per cent for FY23 and FY22, respectively. This indicates that initiatives like PM GatiShakti, Dedicated Freight Corridors (DFC), Bharatmala, and Sagarmala are already driving significant cost optimisation and efficiency gains. Collectively, these developments are thus building a more predictable and cost-competitive industrial environment, which will help India better integrate into GVCs. Further details on initiatives in the infrastructure sub-sectors leading to logistics improvements are presented in the infrastructure chapter (Chapter 9).

Box VIII.9: Why India must strengthen its industrial cluster strategy to compete globally

India's ambition to emerge as a globally competitive industrial hub will critically depend on the strength and scale of its industrial clusters. Evidence from across the world shows that high-performing clusters are not just contributors to industrial activity; they are central to a nation's export growth, attracting foreign investment, driving innovation and enhancing productivity. Countries that have successfully integrated into GVCs, starting from China, Vietnam to South Korea, all have done so through a small number of highly competitive, globally connected clusters that combine economic density with institutional agility. China's Greater Bay Area, for example, generates 35 per cent of national exports and 11 per cent of GDP while occupying less than 1 per cent of land. Vietnam's two key economic regions contribute nearly two-thirds of GDP and trade within just 11 per cent of its land area. These examples highlight a consistent pattern: industrial activity needs concentration, connectivity, and a local environment that grants the freedom to compete and innovate.⁵⁵

Industrial clusters create advantages by bringing firms, suppliers, workers, and logistics together in dense ecosystems. Such co-location increases productivity through shared infrastructure, reduced transaction costs, common labour pools, and continual knowledge spillovers—all of which are essential to industrial competitiveness. A recent success story is Nashik's luggage ecosystem, where Samsonite's facility has emerged as the world's largest by volume, surpassing long-standing European hubs.⁵⁶ The company's decision to further expand locally highlights the importance of established regional supply chains and workforce stability. Such examples demonstrate the potential to 'Make in India for India and the world' by evolving regional clusters into high-productivity global manufacturing bases.

⁵⁴ <https://tinyurl.com/59bsvvey>.

⁵⁵ Inputs from Foundation for Economic Development.

⁵⁶ <https://tinyurl.com/yxffdj3w>.

India has also attempted to build industrial clusters through various schemes over the past few decades, from the 1997 Industrial Park Scheme and the 2005 SEZ Act to the National Industrial Corridor Development Programme and sector-specific clusters for electronics, textiles, and software. While India has several organically developed clusters that bolster domestic production, however, transforming these clusters into globally competitive ecosystems requires addressing two key structural factors. First, achieving optimal scale: the median size of India's clusters is relatively small, often lacking the necessary land area and robust multimodal connectivity essential for global value chain integration. Second, enhancing regulatory flexibility: frameworks governing these zones have yet to fully relax key constraints related to labour, building norms, and ease of doing business, which limit their appeal to international firms seeking speed and predictability.

The gap between domestic strength and global competitiveness underscores the need for an upgraded cluster strategy. Strengthening India's global competitiveness requires re-imagining the cluster model as a high-productivity, reform-enabled ecosystem. The strategic way forward can be built on three pillars. First, it should prioritise scale and location by strategically identifying and anchoring large, high-potential regions in well-connected brownfield locations, ensuring the necessary scale through clear land titling and modern land pooling mechanisms. Second, an empowered institutional mechanism (like the IFSCA at GIFT City) can be granted the authority to ensure regulatory certainty and flexibility, enabling clusters to operate with globally competitive speed. Third, it should involve harnessing private execution by enhancing the role of private developers to masterplan, build, and operate core infrastructure, ensuring market responsiveness and efficiency. Clusters designed under this framework may have the potential to become India's primary engines of growth, accelerating integration into global value chains and supporting the nation's economic growth and resilience.

8.58. Manufacturing activity in India is expanding beyond metropolitan cities to Tier-2 and Tier-3 cities, which possess advantages such as affordable land, lower real estate and wage costs, and proximity to raw material sources. These cities offer significant potential for job creation, sustained economic growth and improved livelihoods, while also supporting decongestion of large metro cities. Sustained investments in connectivity, industrial infrastructure, skilling and supportive policy frameworks can further facilitate many Tier-2 and Tier-3 cities to emerge as competitive manufacturing centers not just domestically but also globally.

8.59. Case studies from smaller industrial centers of Tier-2 and Tier-3 cities illustrate how these advantages are translating into manufacturing competitiveness. For instance, Hubballi-Dharwad-Belagavi is an emerging cluster for advanced manufacturing.⁵⁷ In Belagavi (Karnataka), long-standing foundry and precision-engineering capabilities have evolved into an aerospace-oriented cluster supplying castings and machined components to domestic and global manufacturers, supported by local skills and cluster-

⁵⁷ Source: Karnataka Digital Economy Mission: <https://tinyurl.com/4mwjefxt>.

based industrial development. Similarly, with lower operating costs and improved road, rail and air connectivity with wider supply chains - a diversified manufacturing base anchored in steel processing, metal fabrication and consumer durables has been developed at Hubballi–Dharwad.⁵⁸ India's industrial future thus is likely to be facilitated by its smaller cities across various states.

SCALING UP MSMEs: DEEPENING COMPETITIVENESS, EXPANDING MARKET ACCESS, AND BRIDGING THE CREDIT GAP

8.60. Micro, Small, and Medium Enterprises (MSMEs) form the backbone of India's industrial economy, accounting for approximately 35.4 per cent of manufacturing, around 48.58 per cent of exports, and 31.1 per cent of GDP in the country.⁵⁹ With over 7.47 crore enterprises employing over 32.82 crore persons,⁶⁰ the sector holds its position as the second-largest employer after agriculture. Globally, MSMEs make up about 90 per cent of businesses and are responsible for over 50 per cent of the total global employment.⁶¹ With India's manufacturing sector positioned for greater global integration, MSME sector's role is critical in enabling effective supply-chain participation, fostering local value addition, and supporting inclusive regional growth.

8.61. Despite expanding credit footprints and rising digital integration, access to formal credit remains a binding constraint for many micro-enterprises due to limited collateral and documentation readiness, as is generally the case worldwide. This challenge was also highlighted in the World Bank's Financial Sector Assessment Report for India (2025),⁶² which noted that 27 per cent of MSMEs identify finance as their biggest obstacle. Women-owned MSMEs, in particular, account for a small fraction of commercial credit, though formalisation under Udyam and targeted credit guidelines are gradually addressing this gap. Furthermore, MSME credit has maintained a positive trajectory in recent times, bolstered by several government interventions aimed at enhancing credit flow to the sector.

8.62. MSME credit remained the primary driver of industrial credit growth during H1FY26. In recent periods, the overall MSME credit growth year-on-year (Y-o-Y)

⁵⁸ <https://tinyurl.com/5bmuz77n>.

⁵⁹ Source: National Statistical Office, Ministry of Statistics & Programme Implementation for the year 2023-24. Data for exports is sourced from the portal of Directorate General of Commercial Intelligence and Statistics for the year 2024-25

⁶⁰ Total employment reported by the MSMEs on the Udyam Registration Portal as on 09.01.2026, <https://tinyurl.com/t557udxr>.

⁶¹ NITI Aayog Report on Enhancing MSMEs Competitiveness in India (2025) available at <https://tinyurl.com/2x443e58>.

⁶² World Bank- India Financial Sector Assessment Report (June 2025) <https://tinyurl.com/dv2bsdud>.

has significantly outpaced the Y-o-Y growth observed in large industry credit (Table VIII.3).⁶³ This acceleration can partly be attributed to policy measures, including the revised MSME classification thresholds implemented in April 2025, which expanded the eligibility for priority sector lending, guidelines enabling jewellery-backed small business loans and targeted Budget schemes. Further, NBFCs continued to play a critical role as last-mile credit providers, maintaining strong double-digit lending growth to MSMEs.⁶⁴

Table VIII.3: Growth in deployment of gross bank credit to MSMEs (Y-o-Y per cent), select periods

Segment	March 2024	August 2024	March 2025	August 2025
Micro & Small Enterprises	14.7	13.4	8.8	20.9
Medium Enterprises	13.3	19.2	18.6	13.1
Total MSME Credit	14.3	15.1	11.7	18.5
Large Industry (comparison)	6.4	7.7	6.2	1.8

Source: Monetary Policy Report, RBI Bulletin (October 2025)

8.63. The SME public markets have also witnessed a dramatic expansion over the past two years, driven by buoyant market conditions and digital retail participation. Between 2022–23 and 2024–25, SME IPO listings surged by 87.2 per cent, with the total issue amount growing by 52.7 per cent over the previous year. This boom was characterised by a sharp rise in retail investor participation, facilitated by UPI-based ASBA (Application Supported by Blocked Amount) system, and significant listing gains.⁶⁵ While capital-intensive manufacturing dominated issue sizes, the services sector also saw a rise in listings, indicating the SME segment's enhanced role in mobilising capital for growth.

Box VIII.10: Maharashtra's Pioneering Defence & Aerospace Venture Fund— A State-Led Innovation Finance Model

Maharashtra launched a pioneering Defence & Aerospace Venture Fund (MDAVF), a registered Category II-Alternative Investment Fund (AIF) worth ₹330 crore⁶⁶ in 2018, aimed at catalysing manufacturing capabilities among MSMEs in strategic, high-growth sectors. The Fund, managed by IDBI Capital Markets & Securities, focuses on supporting enterprises engaged in precision engineering, aerospace components, UAV systems, and defence sub-systems. This model blends State capital

⁶³ RBI Bulletin, October 2025: <https://tinyurl.com/2d59t9vx>.

⁶⁴ *ibid*.

⁶⁵ Chattopadhyay, B., & Ganguly, S. (2025, October 20). Fundraising by Indian Small and Medium Enterprises through IPO: Recent Trends and Developments. RBI Bulletin, (October 2025) available at <https://tinyurl.com/bdumbhz8>.

⁶⁶ <https://tinyurl.com/5n97nfcj>.

with professional fund management to provide crucial early-stage and growth-stage support to MSMEs. The Fund invests in companies through various instruments, including equity or equity-linked investments, convertible instruments, and debentures, to achieve risk-adjusted returns.⁶⁷

The Fund has attracted significant private capital into high-technology MSMEs through co-investment and risk-sharing mechanisms and supported MSMEs in acquiring vital defence certifications, facilitating vendor development processes, and achieving their export readiness. The initiative is explicitly aligned with the national indigenisation priorities outlined under the Defence Production and Export Promotion Policy (DPEPP). It leverages strong linkages to Maharashtra's established industrial corridors and emerging defence clusters.

The Fund has extended capital support to a broad set of MSMEs, with investments spanning missile systems, aerospace technologies, naval platforms, and electronic warfare. It has built a diversified portfolio, including investments across multiple enterprises and selective full and partial exits, reflecting a maturing investment cycle. This Fund demonstrates how State-level innovation finance can accelerate MSME technological capabilities in sunrise sectors. Such funds can indeed accelerate MSME participation in strategic value chains by catalysing private investment, facilitating certification and R&D and anchoring enterprises within emerging industrial corridors.

8.64. To strengthen credit linkages, the Credit Guarantee Scheme (CGS) for Micro and Small Enterprises (MSEs) was revamped, effective from 01 April 2023, following a corpus infusion of ₹9,000 crore in the Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE), enhancing the ceiling for guarantee coverage from ₹2 crore to ₹5 crore and reducing the annual guarantee fees to as low as 0.37 per cent. Key enhancements include increasing the extent of guarantee coverage for women-owned MSEs from 85 per cent to 90 per cent. Additionally, effective from 01 April 2025, the government has doubled the ceiling for guarantee coverage from ₹5 crore to ₹10 crore and rationalised the annual guarantee fee for coverage exceeding ₹1 crore.⁶⁸

8.65. The Self-Reliant India (SRI) Fund, launched to infuse ₹50,000 crore as equity funding in MSMEs, has assisted 682 MSMEs by way of investment worth ₹15,442 crore, as of 30 November 2025.⁶⁹ Additionally, the Prime Minister's Employment Generation Programme (PMEGP) assists micro-entrepreneurs by providing margin money subsidies on bank loans. The programme has been expanded to include higher project costs and an enhanced scope of activities. Further, to bolster competitiveness, the MSME Champions Scheme promotes "Zero Defect, Zero Effect" practices through ZED Certification and productivity improvements via the MSME Competitive (Lean)

⁶⁷ <https://tinyurl.com/2b87ffef>.

⁶⁸ Ministry of Micro, Small and Medium Enterprises.

⁶⁹ *ibid*.

Scheme. Innovation is also being institutionalised through the MSME-Innovative component, which facilitates incubation, design interventions, and the protection of IPR.

8.66. Operational challenges such as delayed payments affect liquidity, especially for micro-suppliers. With a view to address these challenges, government's noteworthy interventions include expanding TReDS (Trade Receivables Discounting System) ecosystem and digital invoicing with a reduced monetary limit for onboarding on TReDS for Corporates and CPSEs from ₹500 crore to a turnover of ₹250 crore,⁷⁰ the rapid progress of the ONDC (Open Network for Digital Commerce) ecosystem and the TEAM (Trade Enablement and Marketing Scheme) Initiative (which aims to assist five lakh MSMEs to onboard the platform)⁷¹ offer a transformative path for MSMEs to integrate into formal e-commerce and supply chains at lower transaction costs.

Box VIII.11: Online Dispute Resolution (ODR) for MSMEs: Securing Working Capital

The persistence of delayed payments remains a critical challenge for the MSME sector, with an estimated ₹8.1 lakh crore locked in delayed payments⁷², impacting working capital and restricting growth. When an MSME files a delayed payment case against a buyer, it may strain or even damage the business relationship. Buyers may perceive the filing as an adversarial step and may stop placing new orders or discontinue the partnership altogether. Since MSMEs rely heavily on long-term commercial ties, the fear of losing future business prevent them from pursuing legal option, even when large dues remain pending.

The MSE Scheme on Online Dispute Resolution (ODR) for Delayed Payments, along with the MSME ODR Portal developed under it, introduces a structured process that encourages amicable settlement between seller and buyer before the dispute moves into formal adjudication under the MSMED Act, 2006. This early, dialogue-based resolution mechanism allows MSMEs to recover delayed payments without undermining business relationships.

The portal is designed for accessibility, featuring an end-to-end digital process that is faster and more cost-effective than traditional channels. Its low-cost structure, multi-layered resolution mechanism (combining negotiation, conciliation, and arbitration), and 24X7 availability in multiple languages make dispute resolution economically viable even for very small claims. Ultimately, the ODR initiative enhances trust, promotes contractual discipline, and can directly ease the cash-flow stress that plagues the MSME ecosystem, ensuring greater resilience and competitiveness.

8.67. Overall, the MSME sector is well-positioned to harness the current momentum in the manufacturing sector. In order to continue the momentum, innovative measures

⁷⁰ Source: PIB Release available at <https://tinyurl.com/5z3e7z8h>.

⁷¹ Source: PIB Release available at <https://tinyurl.com/5ce9yyp2>.

⁷² Inputs from Ministry of Micro, Small and Medium Enterprises (MSMEs).

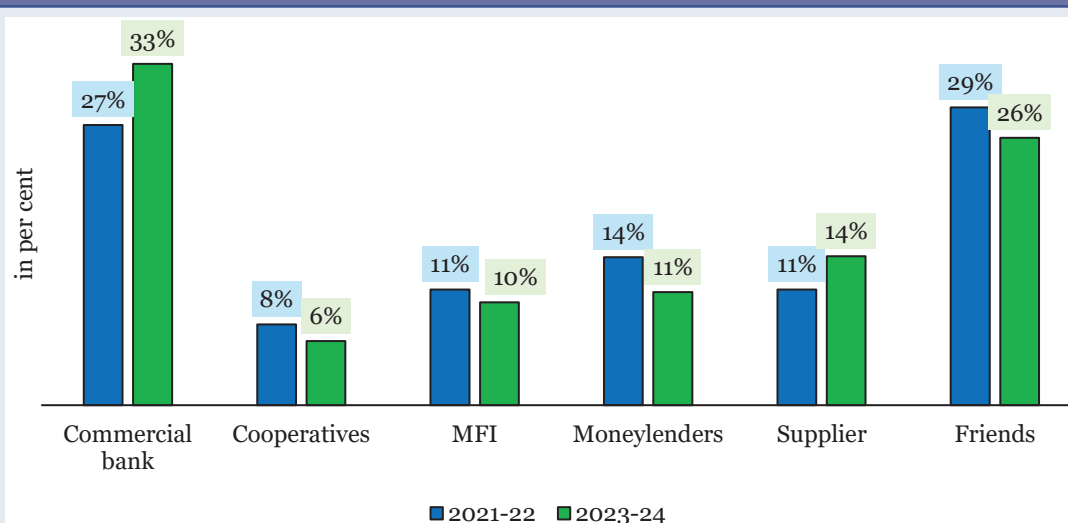
including cash-flow-based lending should be expanded significantly to encourage formal credit access to micro and even first-time borrowers. Further, accelerating digital lending partnerships may also channel timely and affordable finance to a wider base of enterprises.

Box VIII.12: Strengthening Access to Small-Value Credit– Financing Opportunities for Unincorporated Manufacturing Enterprises

Adequate and timely access to credit continues to be a critical enabler of growth for the unincorporated sector. In recent years, access to finance for unincorporated enterprises has improved considerably. According to the Annual Survey of Unincorporated Sector Enterprises (ASUSE), the average outstanding loan per unincorporated non-agricultural establishment increased from ₹50,138 in 2022–23 to ₹53,710 in 2023–24, reflecting a growth of 7.12 per cent.

This expansion in credit availability has also been accompanied by a gradual shift towards formal financing channels, which is a welcome trend. The share of unincorporated manufacturing establishments accessing formal sources of credit rose from 48 per cent in 2021–22 to 51 per cent in 2023–24. Chart VIII.26 further illustrates this transition, showing a 6-percentage-point increase in borrowing from commercial banks between 2021–22 and 2023–24, alongside a 3-percentage-point decline each in reliance on informal lenders. These developments have been possible due to sustained Government efforts to broaden financial inclusion and strengthen the MSME credit ecosystem. Initiatives such as the Pradhan Mantri Jan Dhan Yojana, credit guarantee frameworks, interest subvention schemes, and the MUDRA Yojana have played a significant role in expanding banking outreach and improving access to formal credit for unincorporated enterprises.

Chart VIII.26: Distribution of establishments by Sources of loan outstanding as on last date of reference period

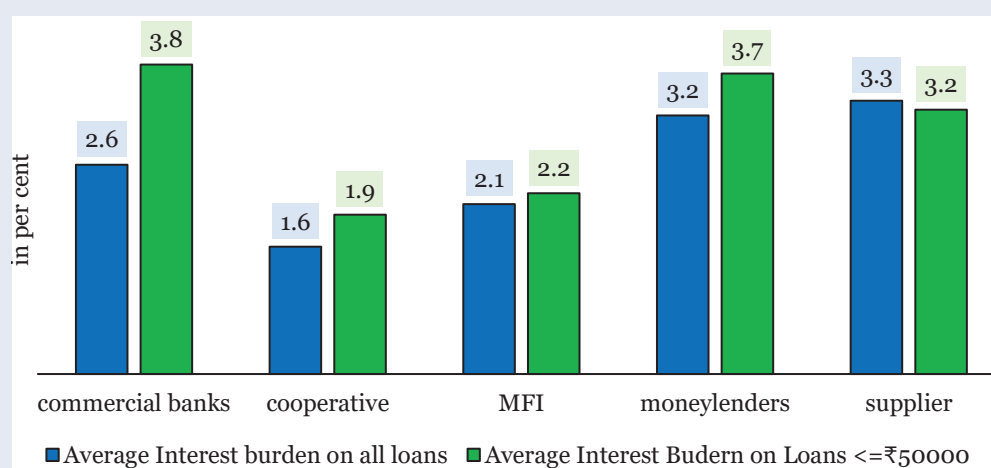


Source: Survey Calculations based on Annual Survey of Unincorporated Sector Enterprises (ASUSE) 2021-22 and 2023-24 Database, MoSPI.

Note: MFI stands for microfinance institution

Further, this shift towards formal sources has also helped reduce the interest burden (measured as interest payable during the reference period as a proportion of outstanding loans) for establishments. As shown in Chart VIII.27, loans obtained from formal institutions typically carry a lower interest burden than those from informal sources. This pattern largely holds true even for small-ticket loans (loan amount less than ₹50,000), except in the case of commercial banks, where the interest burden may be comparable to that of informal lenders, reflecting pricing adjustments for higher perceived risk and lower profitability in such segments. Friends continue to serve as an important financing source owing to strong personal relationships and the availability of low- or zero-interest loans without collateral, particularly for small and urgent funding needs.

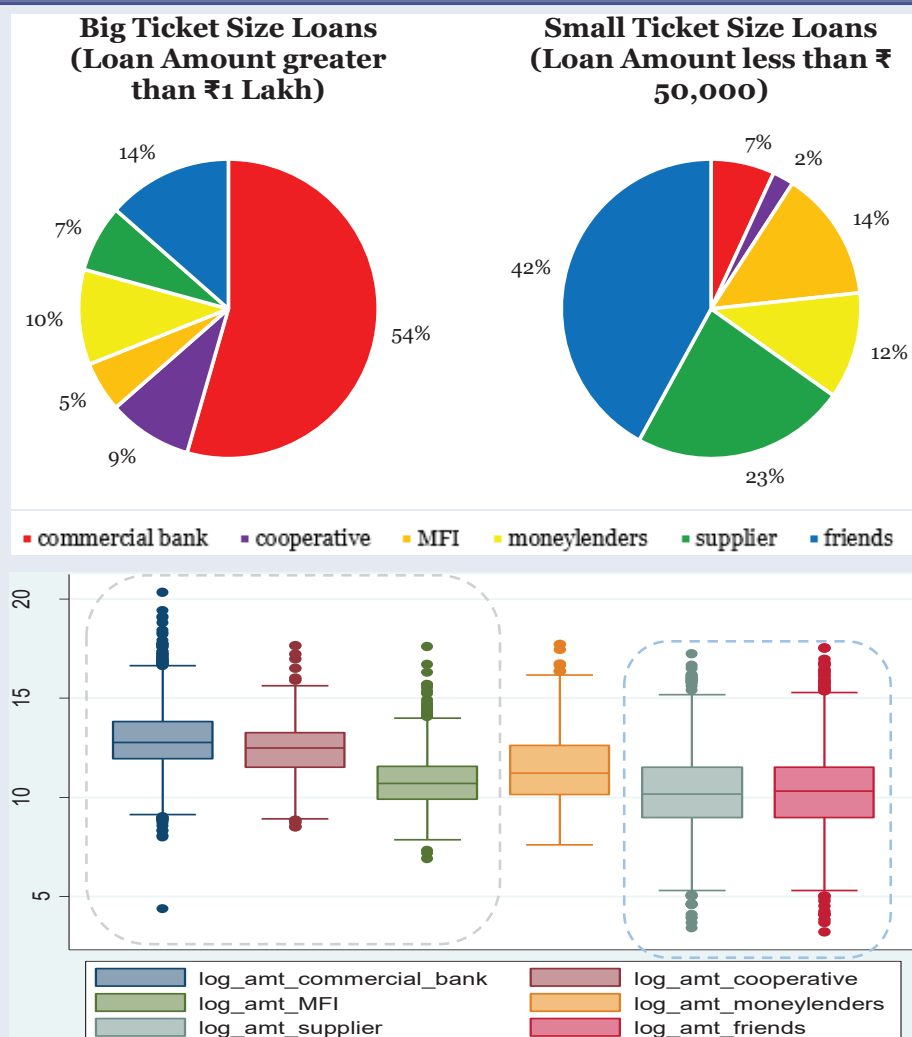
Chart VIII.27: Average interest burden across different sources of finance



Source: Survey Calculations based on Annual Survey of Unincorporated Sector Enterprises (ASUSE) 2023–24 Database, MoSPI.

Chart VIII.28 offers further insights into opportunities for strengthening access to formal finance. While formal institutions have been effective in reducing dependence on informal channels for moderate- and large-sized loan requirements, there is scope for improvement in the provision of formal finance for small-ticket loans. On the demand side, micro and small enterprises often seek small working-capital loans but face barriers such as limited documentation, weak or thin credit histories, the absence of formal banking relationships, and low levels of financial literacy. On the supply side, formal lenders face disincentives in serving this segment due to low profit margins, high transaction and monitoring costs, heightened risk perceptions, and credit products that are often not tailored to the operational needs of these enterprises. Consequently, enterprises continue to rely on informal sources despite their relatively higher costs.⁷³

⁷³ International Finance Corporation (IFC) Report on Financing India's MSMEs: Estimation of Debt Requirement of MSMEs in India, November 2018, available at <https://tinyurl.com/nhp86mp4>.

Chart VIII.28: Distribution of Amount of Loan Outstanding by sources of financing (in per cent)

Source: Survey Calculations based on Annual Survey of Unincorporated Sector Enterprises (ASUSE) 2023–24 Database, MoSPI.

Recognising the distinct credit requirements of unincorporated enterprises, the Government has already initiated several interventions such as Credit Guarantee Schemes for MSMEs,⁷⁴ treatment of bank loans disbursed to MFI under priority sector lending (PSL),⁷⁵ creation of special category of NBFC-MFI (non-deposit taking NBFC which has a minimum of 75 per cent of its total assets deployed towards microfinance loans)⁷⁶ and PSL guidelines, which mandate a sub-target of 7.5 per cent of banks' adjusted net bank credit for micro enterprises.⁷⁷

74 Ministry of Micro, Small & Medium Enterprises, "Credit Guarantee Scheme for Micro & Small Enterprises: Scheme Guidelines," available at <https://tinyurl.com/msc7439y>.

75 Reserve Bank of India (RBI), Master Directions – Priority Sector Lending – Targets and Classification, 2025, FIDD.CO.PSD.BC.13/04.09.001/2024-25 (March 24 2025), available at <https://tinyurl.com/3b2a3s4v>.

76 RBI, <https://tinyurl.com/kb388s9m>

77 RBI, Priority Sector Lending – Targets and Classification, RBI/2014-15/573, FIDD.CO.Plan.BC.54/04.09.01/2014-15 (April 23, 2015), available at <https://tinyurl.com/p5ef8yzk>.

INTEGRATING WITH GLOBAL VALUE CHAINS

8.68. India accounted for an estimated 2.9 per cent of the global manufacturing GVA⁷⁸ and 1.8 per cent of global merchandise exports in 2024,⁷⁹ indicating considerable potential for expanding its global manufacturing footprint. The limited share can be attributed to relatively low participation in GVC trade, particularly on the backward-linked side, where firms import intermediates and components to assemble for export.

8.69. This trend calls for reviewing our policy approach: should India prioritise deepening domestic sourcing or focus on integrating more actively with international production networks? Research supports the latter for a labour-rich economy like India. While greater integration into GVCs, especially through backward participation (importing inputs), initially raises the share of Foreign Value Added (FVA), it generates significantly higher absolute Domestic Value-Added (DVA) and employment in the medium term due to scale effects (Grossman & Rossi-Hansberg, 2008). For a labour-rich economy, export-oriented GVC participation can be a powerful mechanism for employment generation in higher-productivity manufacturing.

8.70. Recent geopolitical realignments and global supply-chain restructuring can also create opportunities for labour-abundant economies to position themselves as competitive assembly and manufacturing hubs in GVCs. In this context, higher import tariffs on intermediates and capital goods relative to final products can lead to inverted duty structures, which raise input costs for domestic manufacturers and discourage assembly and component manufacturing. A renewed and sustained focus on input tariff neutrality may therefore help realise the potential in this window. Towards this end, recent budgets have already initiated correction of inverted duty structures, particularly in several core manufacturing sectors, contributing to a more neutral tariff environment. Going forward, continued calibration of tariffs on intermediates and capital goods—especially in high-growth sectors—can enhance cost competitiveness, deepen assembly and component ecosystems, and support India's emergence as a preferred global production base.

Box VIII.13: Backward GVC participation and Domestic Value Addition: Evidence from India and Vietnam

GVC integration occurs through two principal channels: (i) backward linkages, wherein firms import intermediate inputs for processing and export of final goods, and (ii) forward linkages, where domestically produced inputs are exported for further processing abroad. For late-industrialising economies, empirical literature suggests that backward GVC participation often generates relatively larger early-stage gains, by facilitating entry into

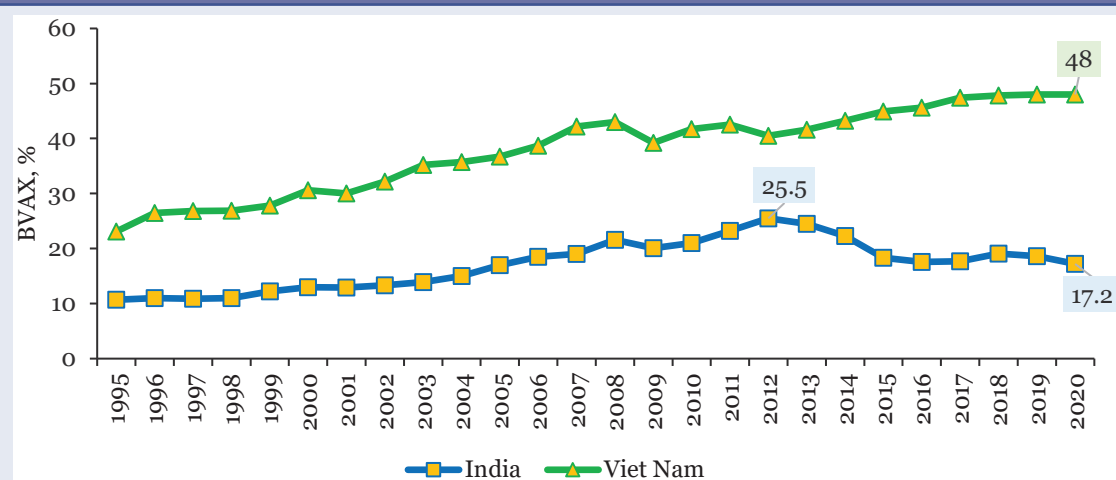
⁷⁸ Based on World Bank data on Manufacturing Value Added (current US\$) (2024) available at <https://tinyurl.com/5ad3mkt2>.

⁷⁹ Based on WTO data (2024) available at <https://tinyurl.com/39d6j6ta>.

assembly-based activities, enabling rapid scale expansion, and supporting employment growth in labour-intensive segments of production (Veeramani & Dhir, 2022).⁸⁰

In the period following trade liberalisation in the 1990s, India experienced a notable increase in backward GVC integration. The BVAX ratio⁸¹ — measured as the share of foreign value added in gross exports — rose from 10.7 per cent in 1995 to 25.5 per cent in 2012, before moderating to 17.2 per cent in 2020. Over a comparable period, Vietnam's BVAX ratio increased more sharply, reaching close to 48 per cent, underpinned by sustained tariff liberalisation, an explicitly export-oriented industrial policy, and deeper integration into global production networks.⁸²

Chart VIII.29: Backward GVC Participation – BVAX (as per cent of Gross Exports), India and Vietnam



Source: OECD-TiVA database.

Cross-country and sectoral evidence indicates that higher backward participation can contribute to increases in total domestic value added (DVA), even when the domestic share per unit of export remains modest in the initial stages. In electronics, for example, Vietnam's BVAX in 2020 was approximately 49.8 per cent — nearly twice India's 25.4 per cent — yet Vietnam generated greater absolute DVA and export value, reflecting the role of scale effects in amplifying domestic value creation. Similar patterns are observed in textiles and related sectors, where Vietnam's stronger GVC integration has been associated with significantly higher export volumes.

These trends suggest that, when supported by competitive input tariffs, efficient logistics and customs processes, predictable regulatory regimes, and an investment-conducive ecosystem, backward-linked GVC integration can function as an important pathway for export growth, employment generation, and gradual strengthening of domestic productive capabilities.

80 Veeramani, C., Dhir, G. Do developing countries gain by participating in global value chains? Evidence from India. *Rev World Econ* 158, 1011–1042 (2022).

81 Backward participation, often measured by BVAX (foreign value added as a share of exports), captures how much imported content is used in a country's exports

82 Veeramani, C., Centre for Development Studies.

For India, a calibrated deepening of participation in global supply chains — particularly in labour-intensive and assembly-linked sectors — represents a potential avenue for advancing job-rich industrialisation within the broader Viksit Bharat@2047 growth trajectory.

Box VIII.14: Why Advanced Manufacturing Matters for India

Advanced manufacturing strategy based on R&D intensive, innovative technology to improve and enhance competitiveness, can contribute not only as a driver of growth, but as a comprehensive ecosystem for enforcing systemic discipline at scale, in a big and late-industrialising economy such as India. Full exposure to global competition, uncompromising quality standards along with thin margins with dynamic delivery standards, significantly limits the scope for regulatory discretion or political access. It would thus encourage manufacturing sector automatically to strictly avoid any compromise on productivity, reliability, and scale. This distinguishes advanced manufacturing from other growth pathways that may raise output without altering firm behaviour. This is also very much in line with India's manufacturing policy based on strategic, selective, and mission-driven, rather than neutral, passive, or purely incentive-based.

The integration of advanced manufacturing with logistics, power, ports, standards, and skills ensures that any institutional weaknesses or deviation quickly and invariably translate into huge economic costs – compelling both the industry or regulator/government to strengthen the weakness or to incur losses or face consequences of poor execution. International experience from Germany, Japan, and South Korea shows that sustained engagement in advanced manufacturing exports has supported both firm-level upgradation and improvements in state capacity.

While not sufficient for inclusive growth and not without risks, it has been observed that the advanced manufacturing remains the pathway that disciplines firms, constrains discretion, and compels governance upgradation at the system level. Other growth drivers—such as infrastructure, capital markets, digital governance, and services exports—can raise incomes, but function best as complements rather than substitutes. Advanced manufacturing matters not because it is aspirational, but because its unforgiving nature forces capability and efficiency at scale on a sustained basis. India's national manufacturing mission strategy in the context of economic development, geopolitical uncertainty, technological disruption, and strategic autonomy can thus accommodate the advanced manufacturing without any frictions.

CONCLUSION: A ROADMAP FOR THE NEXT LEAP

8.71. India's industrial sector continues to display strong momentum despite an evolving and challenging global environment, supported by reforms in infrastructure, logistics, ease of doing business and innovation systems. The rise of medium and high-technology manufacturing, growing traction under PLI schemes, and improving business sentiment point to a strengthening production ecosystem. Domestic

fundraising and credit access have diversified, corporate balance sheets are healthier, and MSMEs are increasingly being integrated into formal supply chains, also through digital platforms and cluster development. India's industrial path is unfolding at a time when the global economy is marked by supply-chain realignments, near-shoring and friend-shoring impulses, and a more fragmented global order. As highlighted in the Economic Survey 2024–25, India must rely more on domestic engines of growth, and this moment represents an inflexion point.

8.72. The next phase of industrialisation will require a calibrated shift from a model centred mainly on import substitution towards one focused on scale, competitiveness, innovation and deeper integration into GVCs. Rather than seeking complete self-reliance in every segment, India needs to build strategic resilience through diversification and creating depth of capabilities. This requires an increase in private sector investment in R&D, technology adoption, skills, and quality systems. MSMEs will be crucial in this journey, evolving from micro-scale production toward deeper participation in formal and export-linked supply chains [See Box VIII.15].

Box VIII.15: Innovation for Strategic Resilience and Indispensability

The global economy has transitioned from an era of efficiency-driven globalisation to one defined by strategic resilience and technological sovereignty. In this landscape, the ability to innovate is no longer merely a driver of productivity; it has become the primary currency of national security and geopolitical leverage.

Despite being the world's fourth-largest economy, India's Gross Expenditure on R&D remains stagnant at ~0.64 per cent of GDP, significantly trailing the 2.5 per cent–5 per cent range of innovation-led economies like the US, China, and Israel. The deficiency in India is not a lack of scientific talent but challenges in collaboration with the private sector and 'translation' at scale.

India excels at Technology Readiness Level (TRL) 1-3, but an industrial economy needs TRL 7-9 products. The intermediate stage - prototyping, piloting, and characterisation - constitutes the stage at which Indian innovation appears to stagnate. To secure a position of leverage in global supply chains, India must continue to work towards bridging the critical gap between academic discovery and industrial commercialisation. Innovation reforms in India must be holistic and systemic, addressing not only research funding but also translation, adoption, and scale.

A critical diagnosis of the Indian ecosystem reveals a persistent market failure where the Indian private sector, historically risk-averse and comfortable with technology licensing or import, has failed to step up as the primary engine of R&D. In this context, the announcement of ₹1 lakh crore RDI Scheme and its operationalization by the Hon'ble Prime Minister in November 2025, reflects a sincere commitment from GOI to collaborate with and incentivize the private sector to undertake innovation and help catapult India towards the frontiers of research, development and innovation.

As we undertake this policy effort to place innovation as a core anchor for India's journey going forward, there is a case for examining the need to complement the approach through the creation of Translational Research Centres (TRCs) that serve as shared national assets for piloting and prototyping. TRCs can be viewed as essential national infrastructure, comparable to physical and digital public infrastructure, and can help reduce the cost and risk of testing, validating, and scaling new technologies for start-ups, MSMEs, industry, and academic institutions. By being built and managed as long-term national assets rather than short-term projects, these centres can provide shared access, continuity, and lasting impact across multiple technology cycles. The ANRF architecture can be leveraged for the identification, establishment, and governance of such TRCs as critical national infrastructure.

The effectiveness of India's innovation reforms depends on adopting a balanced, system-wide approach across the science and technology ecosystem, supported by robust governance frameworks, clear performance metrics, and transparent delivery and accountability mechanisms. In this context, it is essential that the ANRF and the RDI Scheme operate in close coordination to align demand signals, research priorities, institutional incentives, standards-setting, and public procurement. Such alignment will enable the creation of a coherent and integrated national research-to-deployment pipeline, ensuring efficient translation of research outcomes into scalable and deployable technologies.

The data from ASPI's critical technology tracker is a wake-up call – the global technology landscape is bifurcating, and India cannot afford to be a client state. Global capital flows are increasingly gravitating along 'geopolitical faultlines.' Countries that control critical nodes – such as semiconductors, advanced materials, and APIs—possess 'reverse leverage.'

If India fails to ignite this innovation engine, the risks would be long-lasting. India risks remaining a 'service provider' to the developed world, vulnerable to technology denial regimes and supply chain shocks. Conversely, India can choose to work towards the following objective: defensive sovereignty (preventing supply shocks) and offensive leverage (becoming indispensable in high-tech domains). A firm choice shall have to be made.

8.73. As India considers its manufacturing strategy amid economic transformation, geopolitical uncertainty and rapid technological change, manufacturing capacity needs to be viewed as a strategic national asset. It is central to employment, productivity, technological learning, exports and strategic resilience, and therefore warrants a mission-oriented approach beyond purely incentive-based interventions. In this context, government interventions may be considered even in the absence of immediate cost competitiveness for select sectors and technologies of national interest. Policy design can also benefit from treating industrial clusters, rather than standalone firms, as the unit of competition by integrating anchor firms, MSMEs, suppliers, skills, logistics and R&D, with an emphasis on ecosystem depth.

8.74. Sequenced capability building is also critical for strengthening manufacturing competitiveness. This involves supporting progression from assembly to components,

systems and design- or IP-intensive activities through sector-specific capability ladders with time-bound milestones for localisation and value addition. Export performance can serve as an important test of competitiveness, with supported sectors transitioning from import substitution towards exports over defined timelines. While innovative experimentation and learning should be encouraged, persistent stagnation or continued support without capability growth should be avoided. Within this framework, the role of the State extends beyond facilitation to that of strategist, coordinator, risk absorber and capability builder, operating within a federal structure that encourages States also to act as co-owners of national manufacturing priorities aligned with local strengths.

8.75. Strengthening the enabling ecosystem will also be key. Continued tariff rationalisation, especially for intermediates and capital goods, can enhance cost competitiveness and deepen GVC integration. Modern logistics infrastructure, under the PM GatiShakti initiative, regulatory simplification, and a focused industrial cluster strategy—anchored in scale, supplier depth, skills, and shared infrastructure—will be vital for expanding manufacturing beyond existing enclaves,—especially in Tier-2 and Tier-3 cities. The innovation push under ANRF and the RDI Fund, coupled with a future-ready workforce in semiconductors, EVs, green tech and advanced materials, will support technological upgradation. In this context, advanced manufacturing can play a critical role in raising productivity, strengthening export competitiveness and enforcing discipline across firms and create a revamped and robust manufacturing ecosystem.

8.76. Ultimately, a strong industrial sector is a strategic imperative for India. With sustained reform momentum, investment in innovation and human capital, and dynamic private sector participation, the industrial sector can emerge as a central pillar and contribute even more significantly in the journey towards Viksit Bharat@2047.
